

# AN EXPLICIT FINITELY PRESENTED OPERATOR-CORONA NON-MF GROUP

ANONYMOUS

**ABSTRACT.** A countable discrete group is CDE-MF (operator-corona MF) if it embeds into the unitary group of a norm matrix corona. Equivalently, it embeds into the metric quotient of a product of finite-dimensional unitary groups by the sequences converging to the identity in operator norm. We construct an explicit finitely presented group  $E$ , together with a designated central involution  $w \in E$ , such that  $w \neq 1$  but *every* homomorphism from  $E$  to the unitary group of any norm matrix corona sends  $w$  to 1. Thus  $E$  is not operator-corona MF, and neither its full nor its reduced group  $C^*$ -algebra is MF.

The construction starts from  $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ , adjoins an HNN letter implementing  $(v, A) \mapsto (2v, A)$ , and then adjoins one involution. The proof uses coordinatewise averaging and compression, polar correction, a finite-stage property (T) spectral projection, and equality of finite ranks. In general, for the resulting one-sided compression pattern, every finite normal subgroup  $F$  inside the compression-defect normal closure lies in the kernel of every corona representation. A normal property (T) subgroup may replace  $F$ . By contrast, the finite-dimensional argument is elementary and needs neither property (T) nor unitarity.

Consequences include a stably finite non-MF reduced group algebra, a finitely generated sofic non-MF group, permanence failures, scaling families, and structural and marked-space results for the MF radical. An explicit Clifford tower also gives sofic and Connes-embeddable tracial representations in which  $w$  survives. Hence  $w$  belongs to the MF radical but to neither the sofic nor the hyperlinear radical; this does not decide whether  $E$  itself is hyperlinear. Combining the normal-Kazhdan criterion with the torsion-free HNN construction of Fournier-Facio and Hull's acylindrical small cancellation also produces a two-generated, finitely presented, torsion-free, nonsofic property-(T) group whose entire group is its MF radical.

## 1. INTRODUCTION

Let  $(d_n)_{n \in \mathbb{N}}$  be a sequence of positive integers and let

$$(1) \quad \mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$$

be the associated *norm matrix corona*: bounded sequences of matrices modulo sequences converging to 0 in operator norm. Following Blackadar–Kirchberg [BK], a separable  $C^*$ -algebra is *MF* if it embeds into some algebra of the

form (1). We use *MF group* exclusively in the sense introduced by Carrión–Dadarlat–Eckhardt and used by Shulman [CDE, Sh]: a countable discrete group  $E$  is MF if, for some strictly increasing sequence of positive integers  $(d_n)$ , there is an injective homomorphism

$$E \longrightarrow \mathcal{U}(\mathcal{Q}).$$

We shall use the equivalent coordinate model

$$(2) \quad \mathcal{U}_{\text{cor}}((d_n)) := \prod_n \mathcal{U}(M_{d_n}(\mathbb{C})) / \{(u_n) : \|u_n - 1\| \longrightarrow 0\}.$$

The denominator is normal, and polar correction gives a canonical isomorphism

$$(3) \quad \kappa_{(d_n)} : \mathcal{U}_{\text{cor}}((d_n)) \xrightarrow{\cong} \mathcal{U}(\mathcal{Q}),$$

proved in Lemma 6.2. Thus one may equivalently embed  $E$  into  $\mathcal{U}_{\text{cor}}((d_n))$ . For countable groups this is also equivalent, by finite-pair tensor amplification, to operator-norm local models with a uniform positive separation constant. The strict increase of  $(d_n)$  is inessential: place the  $n$ th model in the last block of dimension  $d_0 + \cdots + d_n$  and fill the earlier blocks with identities. This preserves every defect and distance from the identity. Hereafter “MF” means this CDE, or operator-corona, property. It should not be confused with the stronger convention based on reduced-norm convergence of asymptotic representations [Sc, GKEMP], or with strongly PMF, where the approximants are genuine finite-dimensional representations [MdlS, Definition 1.2]. The distinctions matter: for example,  $\text{SL}_4(\mathbb{Z})$  is not strongly PMF but is CDE-MF because it is residually finite. Amenable and residually finite groups are MF, as are free products of amenable groups [TWW, Sh]. The following theorem gives an explicit finitely presented non-MF group.

**Theorem A** (non-MF group). *Let  $\Gamma = \mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$ , let  $\alpha : \Gamma \rightarrow \Gamma$  be the injective endomorphism  $\alpha(v, A) = (2v, A)$ , and let  $a = (e_1, 1) \in \Gamma$ , which does not lie in  $\alpha(\Gamma)$ . Let  $E$  be the finitely presented group obtained from  $\Gamma$  by adjoining a stable letter  $t$  with  $t\gamma t^{-1} = \alpha(\gamma)$  for  $\gamma \in \Gamma$ , an involution  $c$  commuting with  $\Gamma$ , and by imposing that*

$$w := [tct^{-1}, a(tct^{-1})a^{-1}]$$

*is central and satisfies  $w^2 = 1$  (Definition 3.2). Then:*

- (1)  $w \neq 1$  in  $E$ , so  $w$  is a central involution;
- (2) for every sequence  $(d_n)$  and every group homomorphism  $\Theta : E \rightarrow \mathcal{U}_{\text{cor}}((d_n))$  into the corresponding unitary norm corona (2), one has  $\Theta(w) = 1$ .

*In particular  $E$  is not MF, and neither  $C_{\text{max}}^*(E)$  nor  $C_{\text{red}}^*(E)$  is an MF  $C^*$ -algebra.*

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Machine-checking status. The exact displayed presentation, word  $w$ , non-triviality witness, finite-dimensional obstruction, norm-matrix corona, and polar-correction equivalence are formalized. The remaining gap for the literal group is the property-(T) input: the formal development does not yet identify the eight CRW relators with  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$  or certify property (T) for that abstract presentation. A separate Lean theorem proves the same existential conclusion for a noncomputably selected finitely presented Kazhdan-cover witness, not for the displayed group  $E$ . Two interface conventions of the formal development are noted once here: property (T) is formalized as the Kazhdan-pair condition for orthogonal representations on real Hilbert spaces, at fixed universe levels, which is the form the analytic argument consumes; and the corona target is constructed both as the unitary-sequence quotient and as the genuine  $C^*$ -quotient, with the polar-correction isomorphism between them proved and pinned in the audit.

LEAN 4 OTHER WITNESS

The margin labels distinguish exact formalizations, equivalent formulations, supporting lemmas, conditional or partial results, and results for another witness. Only EXACT denotes a one-to-one certification of the printed claim. A statement carrying no margin label has no formal counterpart and rests on the paper proof alone.

A completely explicit presentation of  $E$  — eight generators, with every relation displayed — appears as (8) in Definition 3.2.

The proof is coordinatewise and has five steps. Let  $F \leq H$  be the finite normal subgroup in the general criterion; for Theorem A,  $F = \{1, w\}$ .

- (1) *Finite-group averaging.* From a coordinate model nontrivial on a finite normal subgroup  $F$ , form the coordinate averages  $A_n = |F|^{-1} \sum_{f \in F} U_{f,n}$ . Hermitianize and spectrally round them, then retain the complementary nonzero spectral blocks.
- (2) *Compression and polar repair.* Normality of  $F$  makes the rounded projections asymptotically central. Principal compression therefore preserves multiplication up to vanishing operator-norm error; its Gram defects vanish, and polar correction gives honest unitary matrices without changing the asymptotics (Lemma 7.9).
- (3) *Vanishing finite sum.* On the complementary blocks the Reynolds averages vanish asymptotically, so the repaired model satisfies  $\left\| \sum_{f \in F} V_{f,n} \right\| \rightarrow 0$ .
- (4) *Property-(T) spectral projection.* A finite-stage spectral projection for the adjoint microstates of  $\Gamma$  is almost contained in its  $t$ -translate. The two projections have equal rank, so the complementary block estimate reverses the containment. This forces every generator  $[tct^{-1}, \iota(\lambda)]$  to be Hilbert–Schmidt trivial (Lemma 7.10).
- (5) *Contradiction.* In a tracial matrix ultraproduct the whole normal closure, hence  $F$ , is trivial. A cofinal diagonal extraction makes every  $V_{f,n}$  Hilbert–Schmidt close to  $V_{1,n}$ , contradicting the vanishing finite sum from step (3) (Section 8).

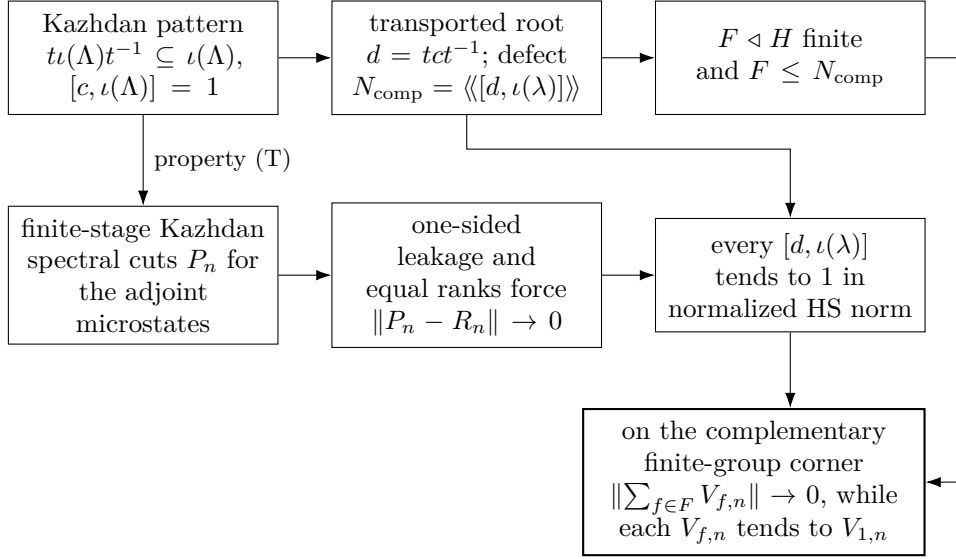


FIGURE 1. The finite-normal obstruction. The upper lane is group-theoretic and the lower lane is the finite-matrix rigidity input. Property (T) supplies a uniform spectral cut, and equality of finite ranks turns its one-sided leakage estimate into two-sided control.

Figure 1 shows where the two halves of the proof meet. The upper lane is purely group-theoretic; the lower lane is the finite-matrix rigidity input. Their common output is incompatible with the complementary finite-group corner.

Nontriviality of  $w$  is witnessed by a completely explicit representation of  $E$  into a semidirect product of a Clifford group by a linear group over  $\mathbb{Z}[1/2]$  (Section 4); no combinatorial theory of HNN extensions is needed.

Two companion results clarify the mechanism. First, in finite dimensions the obstruction is elementary and needs neither property (T), nor unitarity, nor centrality of  $w$ :

For a subgroup  $L \leq H$ , put

$$\mathcal{C}(L) = \{s \in H : sLs^{-1} \subseteq L\}, \quad G_{\text{comp}}(L) = \langle \mathcal{C}(L) \rangle,$$

and define the intrinsic *compression-centralizer defect*

$$(4) \quad \mathfrak{D}(H, L) = \left\langle\left\langle [gzg^{-1}, \ell] : g \in G_{\text{comp}}(L), z \in C_H(L), \ell \in L \right\rangle\right\rangle_H.$$

**Theorem B** (finite-dimensional obstruction). *Let  $k$  be a field,  $V$  a finite-dimensional  $k$ -vector space,  $H$  a group,  $\Gamma \leq H$  a subgroup, and  $t, c \in H$ ,  $a \in \Gamma$  elements with  $t\Gamma t^{-1} \subseteq \Gamma$  and  $c\gamma = \gamma c$  for all  $\gamma \in \Gamma$ . Then every homomorphism  $\pi : H \rightarrow \text{GL}(V)$  satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

More generally, every such  $\pi$  sends the entire subgroup  $\mathfrak{D}(H, \Gamma)$  to the identity.

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Applied to  $H = E$ , Theorem B shows that *every* finite-dimensional representation of  $E$  over *every* field sends  $w$  to the identity: finite-dimensional linear representations do not separate the points of  $E$ , no finite quotient can witness  $w \neq 1$ , and the infinite Clifford witness of Section 4 cannot be replaced by a finite-dimensional one. Property (T) enters the corona argument for one reason: it makes “the projection onto the  $\Gamma$ -fixed space” an element of the corona, uniformly across all coordinates at once.

Second, the fixed-space rigidity in that argument cannot simply be omitted:

**Theorem C** (cyclic-base comparison). *Let  $G_0 = \text{BS}(1, 2) = \langle \gamma_0, t \mid t\gamma_0 t^{-1} = \gamma_0^2 \rangle$ , let  $\Gamma_0 = \langle \gamma_0 \rangle \cong \mathbb{Z}$ , let  $X_0 = G_0/\Gamma_0$ , and let  $E_0 = \text{CLamp}(X_0) \rtimes G_0$  be the corresponding Clifford extension (Construction 4.1), with central sign  $z_0$ . Also define the literal marked-presentation analogue*

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \mid \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right\rangle,$$

where  $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$ . Then  $E_0$  is an amenable MF group and  $z_0 \neq 1$ . There is a quotient homomorphism  $E_{\text{BS}} \twoheadrightarrow E_0$  sending  $w_{\text{BS}}$  to  $z_0$ ; consequently the literal analogous presentation has a corona representation in which its marked involution survives. Nevertheless, every finite-dimensional representation of either group sends the corresponding marked element to the identity.

The formal companion checks the literal seven-relator presentation, its surjective realization onto the subgroup generated by the three displayed images in the concrete Clifford model, survival of the marked word there, and the all-fields finite-dimensional obstruction. It does not package a corona representation of the literal group. In particular, amenability of the exact paper-side group  $E_0$  and the amenability-to-MF implication are not combined into an exact formal counterpart. Thus Theorem C is labelled partial, not exact.

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The quotient statement makes the comparison at the level actually used by the proof: the marked-word relations and witness mechanism survive when the Kazhdan base is replaced by  $\mathbb{Z}$ , but the universal corona conclusion does not. Thus property (T), or a substitute furnishing a uniform fixed-space projection, is essential to this argument; this is not a claim that every possible proof of the concrete Theorem A must invoke property (T).

**Generalizations and consequences.** Theorem A is an instance of a general finite-normal obstruction (Definition 7.1). A marked Kazhdan pattern consists of a property (T) group  $\Lambda$ , a homomorphism  $\iota: \Lambda \rightarrow H$ , and elements  $t, c \in H$  such that  $t\iota(\Lambda)t^{-1} \subseteq \iota(\Lambda)$  and  $[c, \iota(\Lambda)] = 1$ . Theorem 7.2 sends every finite normal subgroup contained in the resulting compression-defect subgroup to the identity under every corona representation; Theorem 7.3

allows a normal property (T) subgroup in place of the finite one. For  $E$ , the relevant subgroup is  $\{1, w\}$ . Corollary 9.5 extends the conclusion to  $C^*$ -subalgebras of norm matrix coronas, while Theorem C shows that the corresponding cyclic pattern need not have the same conclusion.

Section 9 develops structural,  $C^*$ -algebraic, and algorithmic consequences. The principal  $C^*$ -algebraic application is:

**Theorem D** (a stably finite non-MF reduced group algebra).  *$C_{\text{red}}^*(E)$  is a unital separable  $C^*$ -algebra which carries a faithful tracial state — and is therefore stably finite — but is not an MF algebra.*

The faithful trace and matrix-level stable-finiteness argument are formalized for the concrete reduced group algebra. The non-MF conclusion still depends on the unformalized property-(T) endpoint described after Theorem A, so this theorem has no exact badge. The example is nonnuclear and therefore does not affect quasidiagonality results for nuclear stably finite algebras (Remark 9.8).

The same witness also gives failures of quotient and split-extension permanence, sofic and simple envelopes, a functorial MF radical and largest MF quotient, scaling families, infinitely and uncountably many examples, and uniform and marked-space forms of the obstruction. Section 9 states each result and records the additional external input where it is used.

The normal-Kazhdan criterion also removes torsion from the obstruction.

**Theorem E** (torsion-free full MF radical). *There exists a two-generated, finitely presented, torsion-free, acylindrically hyperbolic property-(T) group  $Q$  such that*

$$\text{Rad}_{\text{MF}}(Q) = Q.$$

*The group  $Q$  can be chosen nonsofic. Equivalently, every corona representation of  $Q$ , and every homomorphism from  $Q$  to an MF group, is trivial.*

Theorem E is an unconditional paper theorem: the existence of its small-cancellation quotient is supplied by the cited theorems of Fournier-Facio, Hull, and Osin. The Lean companion verifies the defect saturation and every MF-radical deduction after the corresponding proof-carrying quotient data are supplied; it does not formalize the external theory of acylindrically hyperbolic groups or Hull’s quotient construction. The associated badges in Section 9.11 are therefore supporting declarations, not badges for an exact Lean proof of existence.

**What this paper does not prove.** The proof of Theorem A does *not* show that  $E$  is nonhyperlinear. Proposition 9.15 instead constructs sofic and Connes-embeddable tracial representations in which  $w$  survives. The finite-rank step in the proof is specific to operator norm: nested unitarily equivalent projections need not coincide in the properly infinite operator algebra arising from a tracial ultraproduct. Section 11 explains this failure. The argument also leaves  $E/\langle w \rangle$  unresolved and has no bearing on quasidiagonality of nuclear stably finite algebras.

**External inputs.** The main proof uses property (T) of  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ , the Kazhdan projection, and the Conder–Robertson–Williams presentation [CRW]; Facts 2.1 and 2.2 state the analytic inputs precisely. The Clifford witness and the finite-dimensional obstruction are proved directly. Section 2 records the additional inputs for the cyclic comparison, and Section 9 cites the classical embedding, residual-finiteness, and approximation results at their points of use. Theorem E additionally uses Fournier–Facio’s torsion-free HNN skeleton, Osin’s normal-subgroup lemma, and Hull’s suitable-subgroup and small-cancellation quotient theorems [FF, Osi, Hul].

**Relation to prior work.** The closest conceptual analogue is the recent work on nonsofic groups. One-sided compression of a property (T) subgroup and propagation from its centralizer appear in [OAI, KT], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; related centralizer rigidity for sofic embeddings is proved in [AT]. Those results concern Hamming or tracial approximations. Here coordinatewise averaging, polar correction, and a finite-stage spectral projection produce a specific element in the kernel of every operator-norm corona representation. The Clifford central extension supplies the element whose survival is proved independently.

The scaling base has precedents in work of Cornuier and Abels [Co, Abe], and Thom used a related pattern to construct hyperlinear Kazhdan groups that are not residually finite [Th, Section 2]. The present integral model is  $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$  with  $\alpha(v, A) = (2v, A)$  (Remark 3.5). Its operator-norm argument replaces finite cardinality by the implication

$$\begin{aligned} \text{one-sided leakage of } P_n \text{ into } R_n, \quad \text{rank } P_n = \text{rank } R_n \\ \implies \|P_n - R_n\| \longrightarrow 0. \end{aligned}$$

The complementary finite-group corner then converts this estimate into a fixed nontrivial element sent to the identity by every corona representation.

Slofstra’s constructions supply close precedents for the Clifford group, distinguished central involution, shift, and HNN doubling map [Slo18, Section 2 and Remark 3.3]. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedure in detail [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5]. The present proof combines that local corner procedure with the one-sided Kazhdan compressor; equality of the two finite ranks supplies the reverse projection estimate and hence the universal kernel conclusion.

The analytic passage from operator norm to the Hilbert–Schmidt conjugation representation belongs to the  $\pi \otimes \bar{\pi}$  framework of Bekka and Bekka–Valette [Be, BV]. Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. These are the direct analytic precedents for the conjugation step used here.



Slofstra’s quotients by elements invisible to exact or approximate finite-dimensional representations precede the MF-radical language used below [Slo17, Definitions 2.5–2.7]. Fritz’s marked  $\text{BS}(2, 3)$  example provides additional context for the cyclic comparison [Fri, SV].

Central extensions and rigidity of approximate representations also drive the Frobenius obstruction of De Chiffre–Glebsky–Lubotzky–Thom [DGLT]. In the proof of Proposition 1.6, Bachner–Dogon–Lubotzky treat the finite-central case by a character-isotypic corner, and reduce the general finite-normal case to it by passing to the finite-index centralizer [BDL]. The finite-normal criterion here combines the corresponding coordinate-corner construction with the one-sided Kazhdan compressor and does not assume that the quotient is hyperlinear. Related finite-extension formulations of soficity appear in [GR, Gl].

Carrión–Dadarlat–Eckhardt provide the group-MF framework used here and, through Abels’ group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18]. The cyclic comparison addresses the narrower question of what happens to the marked compression pattern when property (T) is removed.

Finally, failure of the MF property for a group  $C^*$ -algebra does not by itself prevent the group from embedding into a corona: the group property requires only a faithful canonical image in some MF quotient of  $C_{\max}^*(E)$ . The result here produces a specific nontrivial element in the kernel of every such representation. For broader context see [DGLT, BDL, OAI, KT, Sh, Dad, Slo17, Slo18, SV].

**Organization.** Section 2 fixes notation and states the external inputs. Section 3 constructs  $E$ . Section 4 proves  $w \neq 1$  and that  $\Gamma$  embeds in  $E$ . Section 5 proves Theorem B. Section 6 develops the corona lemmas. Section 7 defines marked Kazhdan patterns and their compression-defect subgroups, states the finite-normal obstruction criterion (Theorem 7.2), and develops the coordinate Reynolds and Kazhdan-compression estimates. Section 8 proves the criterion and deduces Theorem A. Section 9 derives Theorems D and E and the further consequences. Section 10 proves Theorem C. Section 11 discusses limitations.

## 2. NOTATION AND EXTERNAL INPUTS

All groups are discrete and countable. All  $C^*$ -homomorphisms and  $C^*$ -algebra embeddings are understood to be unital unless explicitly stated otherwise. For a unital  $C^*$ -algebra  $A$  we write  $\mathcal{U}(A)$  for its unitary group. On  $M_r(\mathbb{C})$  we use the normalized trace  $\text{tr}_r$ , the operator norm  $\|\cdot\|$ , and the normalized Hilbert–Schmidt norm  $\|x\|_2 = \text{tr}_r(x^*x)^{1/2}$ . We will use three elementary facts without further comment:  $\|x\|_2 \leq \|x\|$ ;  $|\text{tr}_r(x)| \leq \|x\|$ ; and  $\|uxv\|_2 = \|x\|_2$  for unitaries  $u, v$ .



Given a sequence  $(d_n)_{n \in \mathbb{N}}$  of positive integers, we write

$$\mathcal{Q}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in \mathbb{N}}^{c_0} M_{d_n}(\mathbb{C}),$$

the quotient of the bounded  $C^*$ -product by the ideal of sequences with  $\|x_n\| \rightarrow 0$ . We call any such algebra a *norm matrix corona*. We also write

$$\mathcal{U}_{\text{cor}}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}} \mathcal{U}(M_{d_n}(\mathbb{C})) / \{(u_n) : \|u_n - 1\| \rightarrow 0\}.$$

A *corona representation* of a group  $E$  is a homomorphism  $E \rightarrow \mathcal{U}(\mathcal{Q})$ . Lemma 6.2 canonically identifies it with a homomorphism into  $\mathcal{U}_{\text{cor}}((d_n))$ , which we call its coordinate model. The group  $E$  is *MF* if it admits an injective corona representation for a strictly increasing dimension sequence. By the dimension-amplification equivalence discussed in the introduction, allowing an arbitrary positive dimension sequence gives the same class of groups.

For a group  $H$ ,  $C_{\max}^*(H)$  denotes the maximal group  $C^*$ -algebra and  $u_h \in C_{\max}^*(H)$  the canonical unitaries. Every unitary representation  $\pi$  of  $H$  on a Hilbert space extends uniquely to a  $*$ -representation of  $C_{\max}^*(H)$ , still denoted  $\pi$ . Every group homomorphism  $\varphi: H_1 \rightarrow H_2$  induces a unital  $*$ -homomorphism  $C_{\max}^*(H_1) \rightarrow C_{\max}^*(H_2)$ ,  $u_h \mapsto u_{\varphi(h)}$ .

We use the commutator convention  $[x, y] = xyx^{-1}y^{-1}$ .

The following two classical facts are the only non-elementary external inputs to the analytic kernel argument of Theorem A.

**Fact 2.1** (property (T); [BHV, Example 1.7.4(i)]).  $\text{SL}_3(\mathbb{Z})$  and  $\Gamma = \mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$  have property (T) [BHV, Example 1.7.4(i), via Corollary 1.4.16 and Theorem 1.7.1]; an explicit-constant proof for  $\text{SL}_n(\mathbb{Z})$ ,  $n \geq 3$ , is [BHV, Theorem 4.2.5]. The group  $\Gamma$  is finitely presented — not as a consequence of property (T), which does not imply finite presentability [BHV, Section 3.4], but by the explicit presentation (6) below, built on [CRW, Theorem 2].

**Fact 2.2** (Kazhdan projection; [AW, Va]). Let  $\Gamma$  be a group with property (T). There is a projection  $p_\Gamma \in C_{\max}^*(\Gamma)$  such that for every unitary representation  $\pi$  of  $\Gamma$  on a Hilbert space  $\mathcal{H}$ , the operator  $\pi(p_\Gamma)$  is the orthogonal projection onto the subspace  $\text{Fix}(\pi) = \{\xi \in \mathcal{H} : \pi(\gamma)\xi = \xi \text{ for all } \gamma \in \Gamma\}$ . Moreover  $u_\gamma p_\Gamma = p_\Gamma u_\gamma = p_\Gamma$  for every  $\gamma \in \Gamma$ .

The identities follow from the defining property: the left identity holds in every representation, and the right identity follows by taking adjoints (or replacing  $\gamma$  by  $\gamma^{-1}$ );  $C_{\max}^*(\Gamma)$  has a faithful representation.

The only additional input, used exclusively in Section 10, is:

**Fact 2.3** ([TWW, Corollary C]). The reduced group  $C^*$ -algebra of a countable amenable group is quasidiagonal.

**Fact 2.4** (quasidiagonal embedding; [BK, Theorem 3.2.2], see also [BO, Chapter 7]). Every separable unital quasidiagonal  $C^*$ -algebra embeds unitaly

into a norm matrix corona. Consequently every countable amenable group is MF.

### 3. THE GROUP

Throughout,  $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$  with multiplication  $(v, A)(w, B) = (v + Aw, AB)$ .

We first record concrete presentation data. By [CRW, Theorem 2],  $\mathrm{SL}_3(\mathbb{Z})$  has the presentation

$$(5) \quad \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

where the generators are realized by the matrices

$$x = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad y = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}.$$

The matrix-labelled presentation is also recorded explicitly in [CLV, Introduction]. In particular, the displayed relations are defining relations, rather than merely relations satisfied by the three matrices. Let  $v_i = (e_i, 1)$  for  $1 \leq i \leq 3$ . Reading off the columns of these matrices gives the following complete finite presentation of the semidirect product:

$$(6) \quad \Gamma = \left\langle v_1, v_2, v_3, x, y, z \mid \begin{array}{l} \text{the eight relations in (5),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle.$$

For completeness, the presentation claim follows directly from the usual semidirect-product normal form. The commuting  $v_i$  define a homomorphism  $V: \mathbb{Z}^3 \rightarrow P$  into the presented group  $P$ , and (5) defines a homomorphism  $B: \mathrm{SL}_3(\mathbb{Z}) \rightarrow P$ . The nine action relations give  $B(A)V(v)B(A)^{-1} = V(Av)$  first for the three displayed generators and the standard basis, hence for every  $A$  and  $v$ . Therefore  $(v, A) \mapsto V(v)B(A)$  is a homomorphism inverse to the evident map from  $P$  onto  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ .

**Lemma 3.1.** *The map  $\alpha(v, A) = (2v, A)$  is an injective endomorphism of  $\Gamma$  with image  $2\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$  of index 8, and  $a := (e_1, 1) \notin \alpha(\Gamma)$ .*

*Proof.* We compute

$$\begin{aligned} \alpha((v, A)(w, B)) &= \alpha(v + Aw, AB) = (2v + 2Aw, AB) \\ &= (2v, A)(2w, B) = \alpha(v, A)\alpha(w, B), \end{aligned}$$

so  $\alpha$  is a homomorphism, clearly injective, with image  $2\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ ; the index equals  $[\mathbb{Z}^3 : 2\mathbb{Z}^3] = 8$ . If  $(e_1, 1) = (2v, A)$  then  $A = 1$  and  $e_1 = 2v$ , which is impossible in  $\mathbb{Z}^3$ .  $\square$

Let  $G$  denote the ascending HNN extension

$$(7) \quad G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \ (\gamma \in \Gamma) \rangle.$$

We will use  $G$  only as an auxiliary group receiving no injectivity claims; every statement we need about it is either a presentation identity or is verified inside an explicit linear quotient (Lemma 4.2).

**Definition 3.2.** Use the six base generators and the relations displayed in (6), and put  $a = v_1$ . Define

$$(8) \quad E = \left\langle v_1, v_2, v_3, x, y, z, t, c \left| \begin{array}{l} \text{all relations in (6),} \\ tv_it^{-1} = v_i^2 \ (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ c^2 = 1, \quad w^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right. \right\rangle,$$

where  $w$  abbreviates the word  $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$ .

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Thus (8) is an explicit presentation with eight generators and a fully displayed finite list of relators. The presentation identities give a canonical homomorphism  $G \rightarrow E$  and a canonical (not a priori injective) homomorphism  $\Gamma \rightarrow E$ ; we write  $\gamma, t, c$  for the images in  $E$ . In  $E$  the defining relations make  $w$  central and impose  $w^2 = 1$ ; Proposition 4.3 below proves that  $w \neq 1$ . The relations  $t\gamma t^{-1} = \alpha(\gamma)$ ,  $[c, \gamma] = 1$  hold for all  $\gamma \in \Gamma$  because they hold on generators. Figure 3 summarizes the three-stage assembly and the fate of the marked word.

Figure 2 isolates the geometry used later in the analytic argument:  $d = tct^{-1}$  centralizes the compressed copy, not necessarily all of  $\Gamma$ , and the marked commutator measures precisely that failure after conjugating  $d$  by  $a$ .

**Remark 3.3** (presentation robustness). Nothing in the proof of Theorem A depends on the particular presentation (6): for *any* finite presentation of  $\Gamma$ , together with words representing the images of the generators under  $\alpha$  and a word representing  $a$ , the analogue of (8) satisfies Theorem A with the same proof, since the argument consumes the relations only through the canonical homomorphisms  $\Gamma \rightarrow E$  and  $G \rightarrow E$ . The role of [CRW, Theorem 2] is therefore confined to certifying that (5) is complete as a presentation of  $\mathrm{SL}_3(\mathbb{Z})$ . The two remaining ingredients of the explicit display are verified independently of [CRW]: the eight relations hold by direct integer computation, and the displayed matrices generate  $\mathrm{SL}_3(\mathbb{Z})$  because explicit words in them realize every elementary transvection — for instance

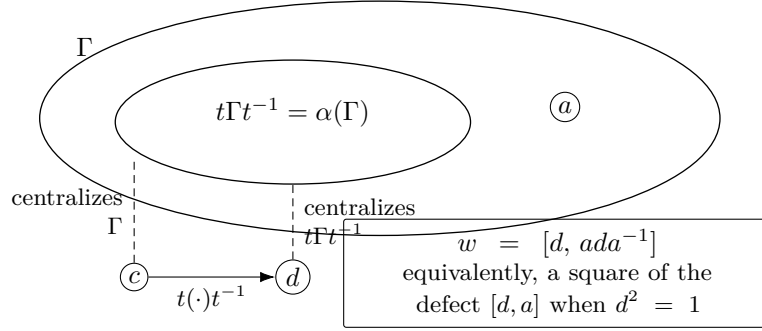


FIGURE 2. The marked-compression pattern. Conjugation by  $t$  carries the base into a proper subcopy; it also carries the centralizer element  $c$  to  $d = tct^{-1}$ , which centralizes that compressed copy. The element  $a \notin t\Gamma t^{-1}$  supplies the marked failure of centrality.

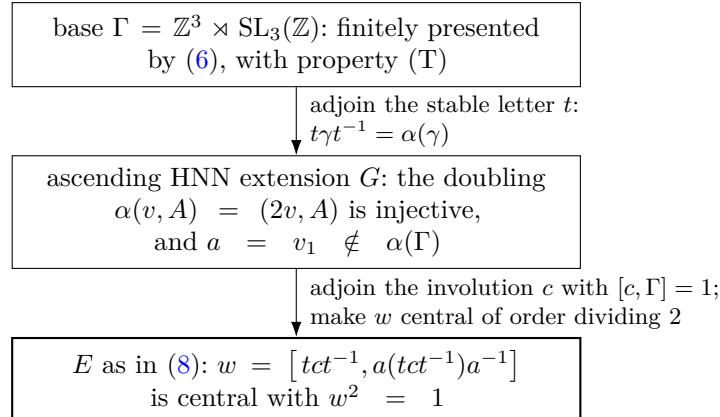


FIGURE 3. Assembly of the group  $E$ . The marked word  $w$  is nontrivial (Proposition 4.3), yet every finite-dimensional representation over every field kills it (Theorem B), and every unitary-norm-corona representation kills it (Theorem A).

$y^{-1}xzy^{-1}x^{-1}$  is the matrix  $1 + E_{12}$ , and the elementary matrices classically generate  $\mathrm{SL}_3(\mathbb{Z})$ .

**Remark 3.4** (nontriviality of the marked image forces proper infinite compression). Let  $\rho$  be any homomorphism of  $E$  to a group, and write  $\Gamma' = \rho(\Gamma)$ . If  $\rho(t)\Gamma'\rho(t)^{-1} = \Gamma'$  — in particular whenever  $\Gamma'$  is finite, since a one-sided inclusion of conjugate finite subgroups is an equality — then  $a' := \rho(t)^{-1}\rho(a)\rho(t) \in \Gamma'$ , so

$$\rho(a(tct^{-1})a^{-1}) = \rho(t)a'\rho(c)a'^{-1}\rho(t)^{-1} = \rho(tct^{-1}),$$

because  $\rho(c)$  commutes with  $\Gamma'$ ; hence  $\rho(w) = 1$ . Consequently any homomorphism with  $\rho(w) \neq 1$  compresses  $\Gamma'$  properly, and  $\Gamma'$  is then infinite. This explains why the pattern is incompatible with standard certificates of the MF property: an infinite property-(T) image rules out amenable targets, and Theorem B rules out linear and residually finite ones.

**Remark 3.5** (why this base). A base for Definition 3.2 must be a finitely presented Kazhdan group with a proper self-embedding. Many familiar rigid groups are co-Hopfian — for example, irreducible lattices in linear semisimple Lie groups in Prasad’s rigidity setting [Pr], and freely indecomposable torsion-free hyperbolic groups by Sela [Se] — so such bases are scarce. In  $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$  the abelian factor exists precisely to break co-Hopfity through the doubling endomorphism, while relative property (T) of the pair  $(\Gamma, \mathbb{Z}^3)$  restores property (T) for the whole group.

#### 4. THE WITNESS: $w \neq 1$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra’s hyperlinear-profile construction [Slo18, Section 2]. The construction below uses the Clifford lamp only as an explicit algebraic witness: the one-sided HNN compression relevant to the obstruction remains in the separate Kazhdan base  $\Gamma$ .

**Construction 4.1** (Clifford groups). Let  $X$  be a set. Define  $\mathrm{Cl}(X)$  to be the quotient of the free unital associative real algebra on symbols  $(c_x)_{x \in X}$  by the relations

$$c_x^2 = 1, \quad c_x c_y = -c_y c_x \quad (x \neq y).$$

For completeness, here is a concrete PBW model. Choose a total order on  $X$ , and let  $A_X$  be the real vector space with basis  $(e_F)$  indexed by the finite subsets  $F \subseteq X$ . Put

$$e_F e_G = (-1)^{\nu(F,G)} e_{F \Delta G}, \quad \nu(F,G) = \#\{(x,y) \in F \times G : y < x\}.$$

Modulo two,  $\nu$  is additive in each variable with respect to symmetric difference, so this multiplication is associative and  $e_\emptyset$  is the identity. The elements  $e_{\{x\}}$  square to 1, and for  $x \neq y$  exactly one of  $\nu(\{x\}, \{y\})$  and  $\nu(\{y\}, \{x\})$  is one; hence they anticommute. The defining relations therefore give a homomorphism  $\mathrm{Cl}(X) \rightarrow A_X$  sending  $c_x$  to  $e_{\{x\}}$ . Conversely the relations reduce every word to an ordered square-free monomial, whose image is  $\pm e_F$ . Thus these monomials are a basis,  $\mathrm{Cl}(X) \neq 0$ , and  $-1 \neq 1$  without invoking a normal-form theorem.

Let  $\mathrm{ClLamp}(X) \leq \mathrm{Cl}(X)^\times$  be the subgroup generated by  $\{\pm 1\} \cup \{c_x : x \in X\}$  and write  $z = -1 \in \mathrm{ClLamp}(X)$ , a central involution. Every permutation of  $X$  induces an algebra automorphism of  $\mathrm{Cl}(X)$  mapping  $c_x \mapsto c_{\sigma(x)}$  and fixing  $z$ ; this gives an action of any group acting on  $X$  by automorphisms of  $\mathrm{ClLamp}(X)$ , and a semidirect product  $\mathrm{ClLamp}(X) \rtimes G$  for any  $G$ -set  $X$ . The group  $\mathrm{ClLamp}(X)$  is the central sign extension of  $\bigoplus_X \mathbb{Z}/2$  defined by the

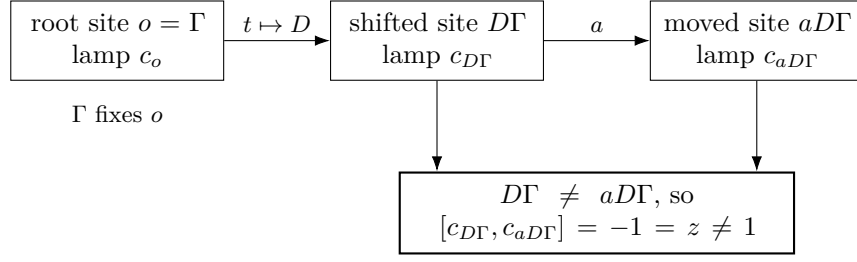


FIGURE 4. The Clifford witness for  $w \neq 1$ . The base fixes the root lamp, while the proper doubling shift creates two distinct Clifford sites whose commutator is the central sign.

ordered-inversion cocycle above; its commutator pairing is  $(-1)^{|F||G|-|F \cap G|}$ . It is a Clifford 2-group of possibly infinite rank, familiar as the multiplicative group generated by Clifford (Majorana) generators in the CAR algebra (see, e.g., [BR2]). The same finite- and infinite-rank Clifford groups with their distinguished central sign appear explicitly in [Slo18, Section 2]; for a modern toolbox of property (T) groups with wreath-like lamp structure see [CIOS].

**Lemma 4.2** (linear model). *Embed  $\Gamma \hookrightarrow \text{GL}_4(\mathbb{Z})$  by  $(v, A) \mapsto \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix}$  and let  $D = \text{diag}(2, 2, 2, 1) \in \text{GL}_4(\mathbb{Z}[1/2])$ . Then  $D\gamma D^{-1} = \alpha(\gamma)$  for all  $\gamma \in \Gamma$ , so  $\bar{G} := \langle \Gamma, D \rangle \leq \text{GL}_4(\mathbb{Z}[1/2])$  is a quotient of  $G$  via  $t \mapsto D$ . In the  $\bar{G}$ -set  $\bar{X} := \bar{G}/\Gamma$  the cosets  $D\Gamma$  and  $aD\Gamma$  are distinct.*

*Proof.*  $D \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} D^{-1} = \begin{pmatrix} A & 2v \\ 0 & 1 \end{pmatrix}$ , which is the matrix of  $\alpha(v, A)$ ; so the assignment  $\gamma \mapsto \gamma$ ,  $t \mapsto D$  respects the relations (7). If  $aD\Gamma = D\Gamma$  then  $D^{-1}aD \in \Gamma$  inside  $\text{GL}_4(\mathbb{Z}[1/2])$ . But  $D^{-1}aD = \begin{pmatrix} 1 & \frac{1}{2}e_1 \\ 0 & 1 \end{pmatrix}$  has a non-integral entry, so it does not lie in the integral group  $\Gamma$ .  $\square$

The three sites used by the witness are summarized in Figure 4. Only the inequality of the two right-hand sites is needed; their Clifford generators then anticommute automatically.

**Proposition 4.3.**  $w \neq 1$  in  $E$ .

*Proof.* Let  $\bar{X} = \bar{G}/\Gamma$  as in Lemma 4.2 and form  $W := \text{CLamp}(\bar{X}) \rtimes \bar{G}$  as in Construction 4.1. Define a map on the generators of (8) by

$$\begin{aligned} g &\mapsto g \in \bar{G} \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto D \in \bar{G}, \quad c \mapsto c_o, \quad o := \Gamma \in \bar{X}. \end{aligned}$$

We check the relations. The base relations in (6) and  $tgt^{-1} = \alpha(g)$  for the six base generators hold in  $\bar{G}$  by Lemma 4.2. We have  $c_o^2 = 1$ . For  $\gamma \in \Gamma$  we have  $\gamma o = o$ , so  $\gamma c_o \gamma^{-1} = c_{\gamma o} = c_o$ , giving the six commutation relations for

c. Under this assignment,

$$tct^{-1} \mapsto c_{D\Gamma}, \quad a(tct^{-1})a^{-1} \mapsto c_{aD\Gamma},$$

and the two sites are distinct by Lemma 4.2, so

$$w \mapsto [c_{D\Gamma}, c_{aD\Gamma}] = c_{D\Gamma}c_{aD\Gamma}c_{D\Gamma}^{-1}c_{aD\Gamma}^{-1} = -1 = z.$$

Since  $z$  is a central involution of  $W$ , the remaining relations  $w^2 = 1$  and  $[w, g] = 1$  for  $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$  are also satisfied. Hence the assignment extends to a group homomorphism  $E \rightarrow W$  sending  $w$  to  $z \neq 1$ , and therefore  $w \neq 1$  in  $E$ .  $\square$

**Remark 4.4.** The target uses the infinite group  $\text{Clamp}(\bar{X})$  when  $\bar{X}$  is infinite, but each group element involves only finitely many Clifford sites; the remaining data are matrices over  $\mathbb{Z}[1/2]$ . No normal-form theory for HNN extensions enters. In particular the consistency of the presentation (8) is verified by an explicit linear-algebraic model.

**Corollary 4.5.** *The canonical homomorphisms  $\Gamma \rightarrow E$  and  $\Gamma \rightarrow G$  are injective; in particular  $E$  contains an isomorphic copy of  $\Gamma$ .*

*Proof.* The homomorphism  $E \rightarrow W$  constructed in the proof of Proposition 4.3 sends each of the six base generators to the corresponding element of  $\bar{G} \leq W$ , so the composite  $\Gamma \rightarrow E \rightarrow W$  agrees on generators with, hence equals, the faithful linear embedding  $\Gamma \hookrightarrow \text{GL}_4(\mathbb{Z}) \leq \bar{G}$  of Lemma 4.2. A homomorphism whose composite with another homomorphism is injective is itself injective; this gives injectivity of  $\Gamma \rightarrow E$ , and of  $\Gamma \rightarrow G$ , since the latter composes with  $G \rightarrow E$  to the former.  $\square$

## 5. THE FINITE-DIMENSIONAL OBSTRUCTION

We prove Theorem B. Everything happens inside the finite-dimensional algebra  $\text{End}_k(V)$ .

*Proof of Theorem B.* Let  $\pi: H \rightarrow \text{GL}(V)$  be a homomorphism and let

$$C = \{x \in \text{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of  $\pi(\Gamma)$ , a linear subspace of  $\text{End}_k(V)$ . Let  $u = \pi(t)$  and consider the linear automorphism  $\Phi(x) = xux^{-1}$  of  $\text{End}_k(V)$ .

We claim  $C \subseteq \Phi(C)$ . Let  $x \in C$  and set  $y = u^{-1}xu$ , so that  $x = \Phi(y)$ . For  $\gamma \in \Gamma$  we have  $t\gamma t^{-1} \in \Gamma$ , hence  $\pi(t\gamma t^{-1})$  commutes with  $x$ , and therefore

$$\pi(\gamma)y = u^{-1}\pi(t\gamma t^{-1})xu = u^{-1}x\pi(t\gamma t^{-1})u = y\pi(\gamma).$$

Thus  $y \in C$ , proving the claim. Since  $\Phi$  is a linear isomorphism,  $\dim_k \Phi(C) = \dim_k C < \infty$ , and the inclusion  $C \subseteq \Phi(C)$  forces

$$(9) \quad C = \Phi(C) = uCu^{-1}.$$



Now  $\pi(c) \in C$  because  $c$  centralizes  $\Gamma$ . By (9),  $\pi(tct^{-1}) = u\pi(c)u^{-1} \in C$ . Since  $a \in \Gamma$ , the element  $\pi(tct^{-1})$  commutes with  $\pi(a)$ , so  $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$ , and therefore

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1. \quad \square$$

The same calculation proves the final, subgroup-valued assertion of Theorem B. The elements of  $H$  whose conjugation action preserves  $C$  in both directions form a subgroup. The argument above puts every one-sided compressor in that subgroup, and therefore puts all of  $G_{\text{comp}}(\Gamma)$  in it. Thus, for  $g \in G_{\text{comp}}(\Gamma)$  and  $z \in C_H(\Gamma)$ , the operator  $\pi(gzg^{-1})$  still belongs to  $C$  and commutes with every  $\pi(\ell)$ ,  $\ell \in \Gamma$ . Hence  $\pi$  kills all the generators in (4), and normality of its kernel kills their normal closure.

**Corollary 5.1.** *The group  $E$  of Definition 3.2 admits no injective homomorphism into  $\text{GL}(V)$  for any finite-dimensional vector space  $V$  over any field. More strongly, finite-dimensional linear representations do not separate its points: they all identify  $w$  with 1. In particular  $E$  is not maximally almost periodic or residually finite,  $C_{\text{max}}^*(E)$  is not residually finite-dimensional as a  $C^*$ -algebra, and no finite quotient of  $E$  can witness  $w \neq 1$ .*

*Proof.* Apply Theorem B with  $H = E$  and the images of  $\Gamma$ ,  $t$ ,  $c$ ,  $a$  in  $E$ ; the hypotheses hold by the defining relations (8). Every finite-dimensional representation kills  $w$ , and  $w \neq 1$  by Proposition 4.3. Moreover the canonical unitary  $u_w$  is not 1 in  $C_{\text{max}}^*(E)$ , because the regular representation sends it to  $\lambda_w \neq 1$ . Every finite-dimensional  $*$ -representation of  $C_{\text{max}}^*(E)$  restricts to a finite-dimensional representation of  $E$  and therefore kills  $u_w - 1$ ; hence  $C_{\text{max}}^*(E)$  is not residually finite-dimensional. A finite quotient of  $E$  gives a finite-dimensional representation after composition with a faithful permutation representation, so it also kills  $w$ .  $\square$

**Remark 5.2.** The proof of Theorem B used only that a finite-dimensional subspace cannot be properly contained in an isomorphic image of itself. An operator-corona model is only asymptotically multiplicative, so its fixed spaces are not literal invariant subspaces. Property (T) repairs this at finite stages: a uniform spectral cut of the adjoint averages captures the almost-fixed vectors. One-sided compression gives a leakage estimate between two equal-rank coordinate projections, and the equal-rank flip supplies the reverse estimate. This is the route used below.

## 6. NORM MATRIX CORONAS

Throughout this section  $\mathcal{Q} = \mathcal{Q}((d_n)_{n \in \mathbb{N}})$  is a norm matrix corona,  $q: \prod_{n \in \mathbb{N}}^{\ell_\infty} M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$  the quotient map.

The following lifting facts are standard perturbation theory; see Loring [Lo] for a systematic treatment.

**Lemma 6.1** (lifting).

- (1) Every projection  $e \in \mathcal{Q}$  lifts to a sequence  $(e_n)_{n \in \mathbb{N}}$  of projections,  $e_n \in M_{d_n}(\mathbb{C})$ .
- (2) Every unitary  $u \in \mathcal{Q}$  lifts to a sequence of unitaries.

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*Proof.* (1) Lift  $e$  to a bounded sequence  $(x_n)$ ; replacing  $x_n$  by  $\frac{1}{2}(x_n + x_n^*)$  and then by the functional-calculus cutoff  $f(x_n)$ , where  $f(t) = \min(1, \max(0, t))$ , we may assume  $0 \leq x_n \leq 1$ . These replacements preserve the quotient class because functional calculus commutes with the quotient map and  $f(e) = e$ . Since  $q((x_n))$  is a projection,  $\|x_n^2 - x_n\| \rightarrow 0$ , so the spectrum of  $x_n$  is eventually contained in  $[0, \frac{1}{3}] \cup (\frac{2}{3}, 1]$ ; for such  $n$  put  $e_n = \chi_{(\frac{1}{2}, \infty)}(x_n)$ , a projection with  $\|e_n - x_n\| \rightarrow 0$ , and set  $e_n = 0$  for the finitely many remaining  $n$ . Then  $(e_n)$  lifts  $e$ .

(2) Lift  $u$  to an arbitrary bounded sequence  $(x_n)$ . Since  $q((x_n))$  is unitary, both  $\|x_n^*x_n - 1\| \rightarrow 0$  and  $\|x_n x_n^* - 1\| \rightarrow 0$ . In particular,  $x_n^*x_n$  is invertible, and hence  $x_n$  is invertible, for all sufficiently large  $n$ . At those indices put  $u_n = x_n(x_n^*x_n)^{-1/2}$ . Then  $u_n$  is unitary, and, once  $\|x_n^*x_n - 1\| \leq \frac{1}{2}$ , continuous functional calculus gives the quantitative estimate

$$\|u_n - x_n\| \leq 2 \|x_n\| \|x_n^*x_n - 1\|.$$

The sequence  $(x_n)$  is bounded, so the right-hand side tends to 0. Setting  $u_n = 1$  at the finitely many remaining indices therefore gives a unitary sequence with  $\|u_n - x_n\| \rightarrow 0$ .  $\square$

**Lemma 6.2** (unitaries in the quotient). *For every sequence  $(d_n)$ , the map*

$$\kappa_{(d_n)}: \mathcal{U}_{\text{cor}}((d_n)) \longrightarrow \mathcal{U}(\mathcal{Q}), \quad [(u_n)] \longmapsto q((u_n)),$$

*is a canonical group isomorphism.*

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*Proof.* The map is well defined, and its kernel consists exactly of the unitary sequences satisfying  $\|u_n - 1\| \rightarrow 0$ , so it is injective. It is surjective by Lemma 6.1(2): every unitary in  $\mathcal{Q}$  has a unitary lift  $(u_n)$ , whose class in (2) maps to it.  $\square$

Stable finiteness of norm matrix coronas is standard and goes back to Blackadar–Kirchberg [BK]; we include the short proofs to keep the external surface explicit.

**Lemma 6.3** (stable finiteness). *Let  $\mathcal{Q}$  be a norm matrix corona.*

- (1) Every isometry in  $\mathcal{Q}$  is a unitary; the same holds in every matrix amplification  $M_k(\mathcal{Q})$ .
- (2) If  $p, q \in \mathcal{Q}$  are projections with  $p \leq q$  and  $v^*v = q$ ,  $vv^* = p$  for some  $v \in \mathcal{Q}$ , then  $p = q$ .

*Proof.* (1) Let  $v \in \mathcal{Q}$  with  $v^*v = 1$  and lift  $v$  to  $(x_n)$  with  $\|x_n^*x_n - 1\| \rightarrow 0$ . Eventually  $x_n$  is invertible (a left-invertible square matrix is invertible), and the unitary  $u_n = x_n(x_n^*x_n)^{-1/2}$  satisfies  $\|u_n - x_n\| \rightarrow 0$ ; hence  $v = q((u_n))$  is unitary. For  $M_k(\mathcal{Q})$  note  $M_k(\mathcal{Q}) \cong \mathcal{Q}((k d_n)_{n \in \mathbb{N}})$ , since matrix amplification commutes with products and with the ideal of null sequences.

(2) This is the standard fact that in a finite algebra a projection is not equivalent to a proper subprojection; we include the short verification for completeness. As in any  $C^*$ -algebra,  $v^*v = q$  gives  $v = vq$  (because  $\|v(1-q)\|^2 = \|(1-q)v^*v(1-q)\| = \|(1-q)q(1-q)\| = 0$ ) and  $vv^* = p$  gives  $v = pv$  (because  $\|(1-p)v\|^2 = \|(1-p)vv^*(1-p)\| = 0$ ). From  $p \leq q$  we get  $qv = q(pv) = pv = v$ , hence  $(1-q)v = 0$  and, taking adjoints,  $v^*(1-q) = 0$  and  $(1-q)v^* = (v(1-q))^* = 0$ . Then  $s := v + (1-q)$  satisfies

$$s^*s = v^*v + v^*(1-q) + (1-q)v + (1-q) = q + 0 + 0 + (1-q) = 1,$$

so  $s$  is an isometry, and

$$ss^* = vv^* + v(1-q) + (1-q)v^* + (1-q) = p + 0 + 0 + (1-q) = p + (1-q).$$

By part (1),  $ss^* = 1$ , i.e.  $p + (1-q) = 1$ , i.e.  $p = q$ .  $\square$

## 7. KAZHDAN COMPRESSION AND THE FINITE-NORMAL CORNER

The corona half of Theorem A will be deduced from a general criterion, whose data we now fix.

**Definition 7.1** (marked Kazhdan pattern). Let  $H$  be a countable group. A *marked Kazhdan pattern* in  $H$  is a tuple  $(\Lambda, \iota, t, c)$ , where  $\Lambda$  is a countable group with property (T),  $\iota: \Lambda \rightarrow H$  is a group homomorphism — *not* assumed injective — and  $t, c \in H$ , subject to

(M1)  $t\iota(\lambda)t^{-1} \in \iota(\Lambda)$  for every  $\lambda \in \Lambda$ ;

(M2)  $c\iota(\lambda) = \iota(\lambda)c$  for every  $\lambda \in \Lambda$ .

Write  $d := tct^{-1}$ . The *compression-defect subgroup* of the pattern is the normal closure

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

There are two canonical enlargements. The integer-orbit defect is

$$N_{\text{orb}} = \langle\langle [t^n ct^{-n}, \iota(\lambda)] : n \in \mathbb{Z}, \lambda \in \Lambda \rangle\rangle_H,$$

and, for  $L = \iota(\Lambda)$ , the intrinsic defect is  $\mathfrak{D}(H, L)$  from (4). Since  $t$  is a one-sided compressor and  $G_{\text{comp}}(L)$  is a group, all powers  $t^n$ , including negative powers, lie in  $G_{\text{comp}}(L)$ . Hence

$$(10) \quad N_{\text{comp}} \leq N_{\text{orb}} \leq \mathfrak{D}(H, L).$$

**Theorem 7.2** (finite-normal obstruction criterion). *Let  $H$  be a countable group with a marked Kazhdan pattern  $(\Lambda, \iota, t, c)$ , and let  $F \trianglelefteq H$  be a finite normal subgroup with  $F \subseteq N_{\text{comp}}$ . Then every corona representation  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(f) = 1$  for all  $f \in F$ .*

**Theorem 7.3** (normal-Kazhdan radical criterion). *Let  $H$  be countable with a marked Kazhdan pattern, and let  $K \trianglelefteq H$  have property (T). If  $K \leq N_{\text{comp}}$ , then*

$$K \leq \text{Rad}_{\text{MF}}(H).$$

*Thus every unitary-norm-corona representation of  $H$  kills  $K$ . In particular, if  $K \neq 1$ , then  $H$  is non-MF. The obstruction subgroup need not be finite, central, or torsion.*

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*Proof.* Fix a homomorphism  $\Theta$  from  $H$  to an operator-norm matrix ultraproduct, and suppose that  $\Theta(k) \neq 1$  for some  $k \in K$ . Replace the target by the countable range  $R = \Theta(H)$ . The subgroup  $K' = \Theta(K) \leq R$  is nontrivial and normal, has property (T), and is contained in the compression defect of the transported pattern.

The inclusion of  $R$  into the original ultraproduct is injective, so the universal MF radical of  $R$  is trivial. The universal-quotient diagonalization therefore gives a faithful representation of  $R$  in one cofinite norm matrix corona. Finally, bounded tensor powers, taken separately for the finitely many ordered pairs in a local test set and then combined by block sum, normalize all pairwise separation constants to 1. Hence  $R$  has a uniformly separated operator-norm matrix approximation.

Apply the normal property (T) corner construction to  $K'$ . In detail: in each coordinate of the uniformly separated model, form the Hermitian average of the adjoint microstates of a finite Kazhdan generating set of  $K'$  and take its high spectral cut, exactly as in Lemma 7.10 but with  $K'$  in place of  $\Lambda$ . Normality of  $K'$  in  $R$  is used here, and only here: conjugation by any element of  $R$  permutes  $K'$ , so the coordinate averages are asymptotically invariant under conjugation by every model unitary, and the resulting spectral projections are asymptotically central, not merely  $K'$ -invariant. Compressing to the complementary block and polar-correcting, as in Lemma 7.9 with the Kazhdan cut in place of the Reynolds average, produces a tracial matrix-ultraproduct representation of  $R$  in which some element of  $K'$  survives. Applied to the same corner model, the compressor argument of Lemma 7.10 makes every generator  $[tct^{-1}, \iota(\lambda)]$  trivial in that tracial ultraproduct; the kernel is normal, so it contains  $N_{\text{comp}}$  and therefore the transported image of  $K$ , hence all of  $K'$  — contradicting the survival above. This is the second Kazhdan projection hidden in the finite-normal argument: it replaces the finite Reynolds average, whose role in Lemma 7.9 was likewise to supply an asymptotically central projection with a nonvanishing complementary block.  $\square$

**Corollary 7.4** (Kazhdan part and exact radical). *Let  $\mathcal{D}_T$  be the subgroup generated by all normal property (T) subgroups of  $H$  contained in  $N_{\text{comp}}$ . Then  $\mathcal{D}_T \leq \text{Rad}_{\text{MF}}(H)$ . If  $N_{\text{comp}}$  itself has property (T), every corona representation factors uniquely through  $H/N_{\text{comp}}$ ; and if this quotient is MF, then*

$$\text{Rad}_{\text{MF}}(H) = N_{\text{comp}}.$$

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**Theorem 7.5** (intrinsic normal-Kazhdan criterion). *Let  $L = \iota(\Lambda) \leq H$  and let  $\mathfrak{D}(H, L)$  be the compression-centralizer defect from (4). If  $K \trianglelefteq H$  has*

property (T) and  $K \leq \mathfrak{D}(H, L)$ , then

$$K \leq \text{Rad}_{\text{MF}}(H).$$

Consequently, if  $\mathfrak{D}(H, L)$  itself has property (T), every corona representation factors through  $H/\mathfrak{D}(H, L)$ ; if that quotient is MF, then

$$\text{Rad}_{\text{MF}}(H) = \mathfrak{D}(H, L).$$

The same containment holds for the subgroup generated by all normal property (T) subgroups contained in  $\mathfrak{D}(H, L)$ .

*Proof.* In the conjugation representation associated to a corona model, let  $P$  be the Kazhdan projection onto the  $L$ -fixed vectors. For every one-sided compressor  $s \in \mathcal{C}(L)$  the fixed-vector calculation gives  $P \leq \beta(s)P\beta(s)^*$ . Stable finiteness upgrades this to equality, so  $P$  commutes with  $\beta(s)$ . It therefore commutes with  $\beta(g)$  for every  $g \in G_{\text{comp}}(L)$ , including inverses and arbitrary finite products.

If  $z \in C_H(L)$ , then the vector represented by  $\Theta(z)$  is  $L$ -fixed. Hence the vector represented by  $\Theta(gzg^{-1})$  is again  $L$ -fixed for every  $g \in G_{\text{comp}}(L)$ . It follows that every generator  $[gzg^{-1}, \ell]$  of (4), and therefore its normal closure, is trivial in the associated tracial matrix ultraproduct. The second Kazhdan-corner argument from Theorem 7.3, now with  $K \leq \mathfrak{D}(H, L)$ , gives the radical containment. The factorization and equality statements follow from the universal property of the quotient and Proposition 9.3.  $\square$

**Theorem 7.6** (faithfully tracial targets). *Let  $A$  be a unital  $C^*$ -algebra carrying a faithful tracial state  $\tau$ , and suppose that  $A \otimes_{\max} A^{\text{op}}$  is stably finite. For every homomorphism  $\Theta: H \rightarrow \mathcal{U}(A)$ ,*

$$\Theta(\mathfrak{D}(H, L)) = 1.$$

*In particular the conclusion holds when  $A$  is nuclear and faithfully tracial. Here  $\tau$  is assumed faithful on all of  $A$ , no separability of  $A$  or countability of  $H$  is needed, and stable finiteness is used only through the statement that an isometry in  $A \otimes_{\max} A^{\text{op}}$  is a unitary. This theorem has no formal counterpart.*

*Proof.* Consider the exact representation

$$\beta(g) = \Theta(g) \otimes \Theta(g)^{*, \text{op}} \quad \text{in } \mathcal{U}(A \otimes_{\max} A^{\text{op}}).$$

On  $L^2(A, \tau)$  — the GNS space of  $\tau$ , on which  $A \otimes_{\max} A^{\text{op}}$  acts by left–right multiplication, using traciality of  $\tau$  for the right action — the action of  $\beta(g)$  is conjugation by  $\Theta(g)$ . The Kazhdan projection  $p_{\Lambda} \in C_{\max}^*(\Lambda)$  of Fact 2.2, pushed through  $\beta \circ \iota$ , therefore projects onto the fixed vectors of this conjugation action. The one-sided compressor calculation of Lemma 9.10 gives, for every  $s \in \mathcal{C}(L)$ , the projection order  $p \leq \beta(s)p\beta(s)^*$  inside  $A \otimes_{\max} A^{\text{op}}$ , where  $p$  denotes the image of  $p_{\Lambda}$ . The isometry construction of Proposition 9.11(1), with  $\beta(s)$  in place of  $u_t$ , exhibits  $\beta(s)p\beta(s)^* - p$  as the defect projection of an isometry; stable finiteness of the maximal tensor product makes that isometry a unitary, so the order becomes equality and  $p$  commutes with  $\beta(s)$ . Thus

the fixed space is invariant under  $G_{\text{comp}}(L)$ . Since  $\Theta(z)$  is fixed whenever  $z \in C_H(L)$ , so is  $\Theta(gzg^{-1})$  for every  $g \in G_{\text{comp}}(L)$ . It commutes in  $L^2(A, \tau)$  with every  $\Theta(\ell)$ ,  $\ell \in L$ ; faithfulness of  $\tau$  makes the GNS representation of  $A$  faithful and so promotes this  $L^2$  equality to equality in  $A$ . Hence all generators of  $\mathfrak{D}(H, L)$  vanish under  $\Theta$ .

If  $A$  is nuclear, maximal and minimal tensor products agree. The product trace on  $A \otimes_{\min} A^{\text{op}}$  is faithful, so this tensor product is stably finite.  $\square$

**Remark 7.7** (higher Kazhdan sectors). The preceding arguments use the isolated trivial representation of  $\Lambda$ . A plausible higher-sector extension replaces it by a finite-dimensional representation  $\rho$  with  $\rho \circ \alpha \simeq \rho$ , where  $\alpha$  is the compression endomorphism. A tuple of lamp vectors transforming as  $\rho$  should be pinned by the corresponding isotypic Kazhdan projection, producing vanishing covariance defects rather than commutators. Brown's finite-dimensional-sector Kazhdan projections [Br] suggest this formulation, but we do not claim the extension as a theorem here: the covariance data and the passage from a one-sided sector inclusion to its precise defect relations still have to be written out.

**Remark 7.8.** Theorem A(2) is the instance  $H = E$ ,  $\Lambda = \Gamma$ ,  $\iota$  the canonical homomorphism, and  $F = \{1, w\}$ ; the membership  $w \in N_{\text{comp}}$  is the identity  $w = [d, \iota(a)]^2$  of Lemma 8.1 below. The property (T) hypothesis cannot simply be deleted: Theorem C realizes the pattern over the base  $\mathbb{Z}$  in an MF group in which the marked involution survives. As with Theorem A, this is not a claim that every possible proof must proceed through property (T); a substitute furnishing the uniform fixed-space projection is what the argument consumes.

The proof uses the following coordinate construction. This formulation is important: the Reynolds average is formed before passing to any quotient, and its complementary spectral corner is repaired coordinate by coordinate.

**Lemma 7.9** (coordinate finite-normal corner). *Let  $F \trianglelefteq H$  be finite, let  $\omega$  be an ultrafilter on an index set  $I$ , and let  $\rho$  be a homomorphism from  $H$  into the operator-norm metric ultraproduct of the groups  $\mathcal{U}(M_{d_i}(\mathbb{C}))$ ,  $i \in I$ . If  $\rho|_F$  is nontrivial, then there are positive integers  $r_n$  and unitaries  $V_{g,n} \in \mathcal{U}(M_{r_n}(\mathbb{C}))$  such that*

$$\|V_{g,n}V_{h,n} - V_{gh,n}\| \longrightarrow 0 \quad (g, h \in H), \quad \left\| \sum_{f \in F} V_{f,n} \right\| \longrightarrow 0.$$

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*Proof.* Choose  $f_0 \in F$  with  $\rho(f_0) \neq 1$ . A standard sequential extraction from the ultraproduct gives representatives  $U_{g,n}$  which are asymptotically multiplicative in operator norm and a constant  $\delta > 0$  such that

$$\|U_{f_0,n} - U_{1,n}\| \geq \delta$$

for every  $n$ . We now construct the corner model from these coordinates. Put

$$A_n = \frac{1}{|F|} \sum_{f \in F} U_{f,n}, \quad H_n = \frac{1}{2}(A_n + A_n^*).$$

Left and right translation permute  $F$ , while conjugation by any  $g \in H$  permutes  $F$  by normality. Expanding the finitely many multiplicative defects therefore shows, in operator norm, that

$$A_n^2 - A_n \longrightarrow 0, \quad A_n - H_n \longrightarrow 0, \quad U_{g,n} A_n U_{g,n}^* - A_n \longrightarrow 0.$$

Let  $P_n = \chi_{(1/2, \infty)}(H_n)$  and  $Q_n = 1 - P_n$ . Spectral rounding gives

$$\|P_n - H_n\| \leq 2 \|H_n^2 - H_n\| \longrightarrow 0,$$

so  $P_n - A_n \rightarrow 0$  and  $U_{g,n} P_n U_{g,n}^* - P_n \rightarrow 0$  for every fixed  $g$ . The separation of  $f_0$  forces  $Q_n \neq 0$  eventually: if  $Q_n = 0$ , then  $P_n = 1$ , and the relations  $U_{f_0,n} A_n - A_n \rightarrow 0$ ,  $P_n - A_n \rightarrow 0$ , and  $U_{1,n} - 1 \rightarrow 0$  would imply  $\|U_{f_0,n} - U_{1,n}\| < \delta$ .

Conjugate into an eigenbasis of  $H_n$  and let  $C_{g,n}$  be the principal block of  $U_{g,n}$  on  $Q_n \mathbb{C}^{d_n}$ ; put  $r_n = \text{rank } Q_n > 0$ . The preceding commutator estimate makes both off-diagonal blocks tend to zero. Hence

$$\|C_{g,n} C_{h,n} - C_{gh,n}\| \longrightarrow 0, \quad \|C_{g,n}^* C_{g,n} - 1\| \longrightarrow 0.$$

For each fixed  $g$  and all sufficiently large  $n$  the compressed Gram matrix is invertible; take the polar correction

$$V_{g,n} = C_{g,n} (C_{g,n}^* C_{g,n})^{-1/2}.$$

When the Gram defect is at most  $1/2$ , functional calculus gives the dimension-free estimate

$$\|V_{g,n} - C_{g,n}\| \leq 2 \|C_{g,n}^* C_{g,n} - 1\|,$$

and at the remaining coordinates set  $V_{g,n} = 1$ . Thus polar correction is asymptotically invisible and the  $V_{g,n}$  remain asymptotically multiplicative.

Finally, the normalized sum of the uncorrected corner matrices is precisely the  $Q_n$ -block of  $A_n$ . Since  $P_n - A_n \rightarrow 0$  and  $Q_n P_n = 0$ ,

$$\left\| \sum_{f \in F} C_{f,n} \right\| \longrightarrow 0.$$

There are only finitely many  $f \in F$ , so replacing every  $C_{f,n}$  by its polar correction preserves this limit and proves the lemma.  $\square$

The second ingredient is the finite-stage Kazhdan-compression calculation.

**Lemma 7.10** (compression defects collapse). *Let  $(V_{g,n})$  be any operator-norm asymptotic unitary representation of  $H$ . For a marked Kazhdan pattern  $(\Lambda, \iota, t, c)$ , with  $d = tct^{-1}$ , one has*

$$\|V_{[d, \iota(\lambda)], n} - V_{1,n}\|_2 \longrightarrow 0 \quad (\lambda \in \Lambda).$$

*Proof.* Apply the adjoint microstates  $\text{Ad } V_{\iota(\lambda),n}$  to  $L^2(M_{r_n}, \text{tr}_{r_n})$ . Choose a finite symmetric Kazhdan set  $S \subseteq \Lambda$ , with  $1 \in S$ , generating  $\Lambda$ , and a Kazhdan constant  $0 < \kappa \leq 1$ . Fix

$$1 - \frac{\kappa^2}{4|S|} < \theta < 1,$$

let  $H_n$  be the Hermitian average of the adjoint  $S$ -microstates, and put  $P_n = \chi_{(\theta, \infty)}(H_n)$ . Spectral capture gives, for every fixed  $\lambda \in \Lambda$ ,

$$\|(\text{Ad } V_{\iota(\lambda),n} - 1)P_n\| \longrightarrow 0.$$

The constants can be fixed uniformly in the matrix size. Given a requested squared Hilbert–Schmidt error  $\varepsilon > 0$ , set

$$\eta = \frac{(1 - \theta)\varepsilon}{192}.$$

At a sufficiently large common stage, each of the two conjugation defects of  $V_{c,n}$  associated to every  $s \in S$  is at most  $\eta$  in operator norm. The Hermitian average therefore has residual norm at most  $\eta$ , and the spectral-cut inequality gives the normalized root-capture estimate

$$(11) \quad \frac{1}{r_n} \|(1 - P_n) \text{vec}(V_{c,n})\|_2^2 \leq \frac{\varepsilon}{192}.$$

For each generator in the Kazhdan set, (M1) supplies a  $\lambda' \in \Lambda$  with  $t\iota(\lambda)t^{-1} = \iota(\lambda')$ . The multiplicative defects then show that the rotated corner  $V_{t,n}^* P_n$  has asymptotically no leakage below the same spectral cut. If  $R_n = (\text{Ad } V_{t,n})P_n(\text{Ad } V_{t,n})^*$ , this gives

$$\|(1 - R_n)P_n\| \longrightarrow 0.$$

They have equal rank at every coordinate. The finite-dimensional projection flip, applied with  $\varepsilon' = \min\{1/2, \varepsilon/2\}$ , bounds the reverse leakage by  $\varepsilon'/\sqrt{1 - (\varepsilon')^2} \leq 2\varepsilon' \leq \varepsilon$ . Thus

$$\|(1 - P_n)R_n\| \longrightarrow 0,$$

and hence

$$\|P_n - R_n\| \longrightarrow 0.$$

This equal-rank projection argument is the finite-dimensional replacement for the stable-finiteness step one might be tempted to perform only after taking a  $C^*$ -quotient; Figure 5 depicts the geometry.

Because  $c$  centralizes  $\iota(\Lambda)$ , the vector represented by  $V_{c,n}$  is asymptotically captured by  $P_n$ . The equality just obtained transports that capture to  $V_{d,n}$ ; hence conjugation by  $V_{\iota(\lambda),n}$  moves  $V_{d,n}$  by a quantity tending to zero in normalized Hilbert–Schmidt norm. Unitary invariance identifies this quantity with  $\|V_{[d, \iota(\lambda)],n} - V_{1,n}\|_2$ , up to the vanishing multiplicative defects. More explicitly, the finite-stage vector chain gives

$$(12) \quad \text{dist}_{2,n}([d, \iota(\lambda)], 1)^2 \leq 8\delta_n^2 + 64\ell_n^2 + 64\rho_n,$$



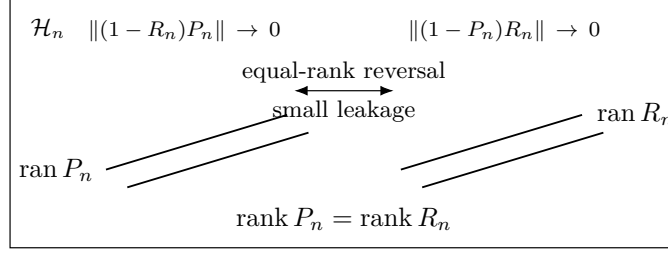


FIGURE 5. Finite-dimensional reversal. One-sided spectral leakage and equality of ranks control the complementary block in the reverse direction; together the two estimates force  $\|P_n - R_n\| \rightarrow 0$ . The drawn planes are schematic ranges inside the common finite-dimensional space  $\mathcal{H}_n$ .

where  $\delta_n$  is the fixed-element Kazhdan displacement,  $\ell_n$  is the reverse compressor leakage, and  $\rho_n$  is the normalized mass in (11). Choose their eventual bounds as

$$\delta_n \leq \sqrt{\varepsilon/24}, \quad \ell_n \leq \sqrt{\varepsilon/192}, \quad \rho_n \leq \varepsilon/192.$$

Then the right side of (12) is at most

$$8\frac{\varepsilon}{24} + 64\frac{\varepsilon}{192} + 64\frac{\varepsilon}{192} = \varepsilon.$$

This proves the claim with constants independent of  $r_n$ . These are the same 8/64/64 finite-stage estimates used by the formal proof; no passage to an abstract limiting projection is hidden here.  $\square$

## 8. PROOF OF THE CRITERION AND OF THEOREM A

*Proof of Theorem 7.2.* Suppose, for contradiction, that some corona representation  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  is nontrivial on  $F$ . Compose with  $\kappa_{(d_n)}^{-1}$  to obtain its coordinate model. For some  $f_0 \in F$  the distance from its image to 1 has positive limit superior. Pass to a subsequence on which that distance is uniformly positive and then to any free ultrafilter on the subsequence. Lemma 7.9 produces an operator-norm asymptotic unitary representation  $(V_{g,n})$  such that

$$(13) \quad \left\| \sum_{f \in F} V_{f,n} \right\| \longrightarrow 0.$$

Regard the same matrices as an asymptotic representation for the normalized Hilbert–Schmidt metric, and fix a free ultrafilter  $\omega$  containing the cofinite filter. They define a homomorphism  $\rho_2$  into the corresponding tracial unitary ultraproduct. By Lemma 7.10,

$$\rho_2([d, \iota(\lambda)]) = 1 \quad (\lambda \in \Lambda).$$

The kernel of a homomorphism is normal; hence it contains  $N_{\text{comp}}$ , and therefore  $F$ . Consequently, for every  $f \in F$ ,

$$(14) \quad \lim_{n \rightarrow \omega} \|V_{f,n} - V_{1,n}\|_2 = 0.$$

To return from the ultrafilter statement to an ordinary sequence, put  $\eta_k = 1/(k+1)$ . Since  $F$  is finite, for every  $k$  the set of indices  $n \geq k$  for which

$$\|V_{f,n} - V_{1,n}\|_2^2 < \eta_k \quad (f \in F)$$

belongs to  $\omega$  and is therefore nonempty. Choose one such index  $\phi(k) \geq k$  and reindex by  $W_{g,k} = V_{g,\phi(k)}$ . Cofinality of  $\phi$  preserves asymptotic multiplicativity and (13), while

$$\|W_{f,k} - W_{1,k}\|_2 \rightarrow 0 \quad (f \in F).$$

Because  $F$  is finite,

$$\left\| \sum_{f \in F} W_{f,k} - |F|W_{1,k} \right\|_2 \leq \sum_{f \in F} \|W_{f,k} - W_{1,k}\|_2 \rightarrow 0.$$

On the other hand,  $W_{1,k}$  is unitary, so  $\| |F|W_{1,k} \|_2 = |F|$ , whereas

$$\left\| \sum_{f \in F} W_{f,k} \right\|_2 \leq \left\| \sum_{f \in F} W_{f,k} \right\| \rightarrow 0$$

by (13) after reindexing. The triangle inequality would force  $|F| = 0$ , a contradiction. Thus every corona representation kills  $F$ .  $\square$

**Lemma 8.1** (the marked word is a defect square). *Let  $H_1$  be a group and  $d, a_1 \in H_1$  with  $d^2 = 1$ . Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

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*Proof.* Put  $b = a_1 d a_1^{-1}$ , so  $b^2 = a_1 d^2 a_1^{-1} = 1$ ; thus  $b^{-1} = b$  and  $d^{-1} = d$ . Then  $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d (a_1 d a_1^{-1}) = db$ , and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

*Proof of Theorem A.* Part (1) is Proposition 4.3. For part (2), let  $\iota: \Gamma \rightarrow E$  be the canonical homomorphism and  $a = v_1 = (e_1, 1)$ . The tuple  $(\Gamma, \iota, t, c)$  is a marked Kazhdan pattern in  $E$ :  $\Gamma$  has property (T) by Fact 2.1; (M1) holds because  $t \iota(\gamma) t^{-1} = \iota(\alpha(\gamma))$  for every  $\gamma \in \Gamma$  (both sides are homomorphisms in  $\gamma$ , and (8) gives the identity on the six base generators); (M2) holds because  $[c, g] = 1$  in (8) for the six base generators  $g$ . In  $E$  we have  $d^2 = (t c t^{-1})^2 = t c^2 t^{-1} = 1$ , so Lemma 8.1 with  $a_1 = \iota(a)$  gives

$$w = [d, \iota(a) d \iota(a)^{-1}] = [d, \iota(a)]^2 \in N_{\text{comp}},$$

since  $[d, \iota(a)]$  is one of the defining generators of  $N_{\text{comp}}$ . The subgroup  $F = \{1, w\} = \langle w \rangle$  is finite and normal: centrality and  $w^2 = 1$  are defining relations,

while Proposition 4.3 gives  $w \neq 1$ . Moreover  $F \subseteq N_{\text{comp}}$ . Theorem 7.2 therefore kills  $w$  after composing any  $\Theta: E \rightarrow \mathcal{U}_{\text{cor}}((d_n))$  with the canonical isomorphism  $\kappa_{(d_n)}$  of Lemma 6.2. Since  $\kappa_{(d_n)}$  is injective,  $\Theta(w) = 1$ . This is part (2).

For the final assertions of Theorem A: an MF embedding of  $E$  would be an injective corona representation, which is impossible because every corona representation sends the nontrivial element  $w$  to 1. If  $C_{\text{red}}^*(E)$  were an MF algebra, a unital  $C^*$ -embedding  $C_{\text{red}}^*(E) \hookrightarrow \mathcal{Q}$  composed with the canonical map  $E \rightarrow \mathcal{U}(C_{\text{red}}^*(E))$ , and then with  $\kappa_{(d_n)}^{-1}$ , would exhibit an injective corona representation of  $E$ : the map  $g \mapsto \lambda_g$  is injective, although the quotient map  $C_{\text{max}}^*(E) \rightarrow C_{\text{red}}^*(E)$  need not be faithful. The canonical map  $E \rightarrow \mathcal{U}(C_{\text{max}}^*(E))$  is also injective, since its composition with the regular representation is  $g \mapsto \lambda_g$ ; after the same  $\kappa_{(d_n)}^{-1}$  conversion, this treats  $C_{\text{max}}^*(E)$ .  $\square$

**Remark 8.2.** The proof shows more than non-injectivity of each individual  $\Theta$ : the fixed element  $w$  lies in the kernel of *every* corona representation of  $E$ , so every corona representation factors through  $E/\langle w \rangle$  (note  $\langle w \rangle = \{1, w\}$  is central, hence normal). In the language of approximation radicals,  $w$  is a nontrivial element of the “MF radical” of  $E$ . This terminology is meant in analogy with Slofstra’s earlier quotients  $G^{\text{fin}}$  and  $G^{\text{fa}}$ , obtained by killing elements invisible to exact and approximate finite-dimensional representations, respectively [Slo17, Definitions 2.5–2.7]. By Theorem 7.2 the same holds in every countable group with a marked Kazhdan pattern: every finite normal subgroup of the compression-defect subgroup lies in the MF radical. Section 9 develops the radical systematically; we do not know whether  $E/\langle w \rangle$  is MF, i.e. whether the MF radical of  $E$  is exactly  $\{1, w\}$ .

**Remark 8.3.** No lower bound on the corner ranks  $r_n$ , or on their density in  $d_n$ , is used anywhere: Lemma 7.9 equips each nonzero complementary block with its own normalized trace. Thus all subsequent Hilbert–Schmidt estimates are relative to that block. This is what makes the operator-norm setting insensitive to small relative ranks, which block the same argument in Hilbert–Schmidt metrics (Section 11).

## 9. CONSEQUENCES

Throughout this section, “corona representation” and “MF” are as in Section 2.

The discussion has three parts. First, the MF radical packages the universal kernel and extends the obstruction to other targets. Second, the explicit group yields  $C^*$ -algebraic and permanence consequences. Third, envelope, scaling, multiplicity, recognition, and marked-space results propagate the same witness without changing the analytic argument.

### 9.1. The MF radical.

**Definition 9.1.** For a countable group  $G_1$ , the *MF radical* is

$$\text{Rad}_{\text{MF}}(G_1) = \bigcap \{ \ker \Theta : \Theta \text{ a corona representation of } G_1 \},$$

a normal subgroup of  $G_1$ . This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the approximation kernels studied in [NST].

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**Lemma 9.2** (portability). *For every homomorphism  $\varphi: G_1 \rightarrow G_2$  of countable groups,*

$$\varphi(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_2) :$$

*the MF radical is functorial. In particular, if  $\varphi: E \rightarrow G_2$  satisfies  $\varphi(w) \neq 1$ , then  $G_2$  is not MF.*

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*Proof.* If  $x \in \text{Rad}_{\text{MF}}(G_1)$  and  $\Theta$  is a corona representation of  $G_2$ , then  $\Theta \circ \varphi$  is a corona representation of  $G_1$ , so  $\Theta(\varphi(x)) = 1$ . For the second statement:  $w \in \text{Rad}_{\text{MF}}(E)$  by Theorem A, so  $1 \neq \varphi(w) \in \text{Rad}_{\text{MF}}(G_2)$ , and an injective corona representation of  $G_2$  would kill  $\varphi(w)$ , a contradiction.  $\square$

**Proposition 9.3** (the largest MF quotient). *Let  $G_1$  be a countable group and  $R = \text{Rad}_{\text{MF}}(G_1)$ .*

- (1)  *$R$  is the kernel of a single corona representation of  $G_1$ .*
- (2)  *$G_1/R$  is MF.*
- (3) *Every homomorphism from  $G_1$  to an MF group factors through  $G_1/R$ .*

*In particular  $G_1$  is MF if and only if  $R = \{1\}$ . The statement is presumably folklore once the radical is defined; we include a proof for completeness and claim no novelty for it.*

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*Proof.* (3) If  $\varphi: G_1 \rightarrow M$  with  $M$  MF and  $j: M \rightarrow \text{U}_{\text{cor}}((d_n))$  is an injective corona representation, then  $j \circ \varphi$  is a corona representation of  $G_1$ , so  $j(\varphi(R)) = 1$  and, by injectivity of  $j$ ,  $\varphi(R) = 1$ .

(1) and (2). If  $R = G_1$ , the trivial representation has kernel  $R$  and  $G_1/R$  is trivial, hence MF; so assume  $R \neq G_1$  and let  $(g_i)_{i \geq 1}$  list, possibly with repetitions, elements of  $G_1 \setminus R$  meeting every nontrivial coset of  $R$ . For each  $i$  choose a corona representation and write it, through  $\kappa_{(d_n^{(i)})}$ , as  $\Theta_i: G_1 \rightarrow \mathcal{U}(\mathcal{Q}_i)$ ,  $\mathcal{Q}_i = \mathcal{Q}((d_n^{(i)})_{n \in \mathbb{N}})$ , with  $\Theta_i(g_i) \neq 1$ ; put  $\varepsilon_i = \frac{1}{2} \|\Theta_i(g_i) - 1\| > 0$ , and choose unitary lifts  $(U_{g,n}^{(i)})_{n \in \mathbb{N}}$  of  $\Theta_i(g)$  for every  $g \in G_1$  (Lemma 6.1), with  $U_{1,n}^{(i)} = 1$ . Recall that the quotient norm in a corona is the limit superior of the coordinate norms: subtracting a null sequence does not change the limit superior, and truncating finitely many coordinates realizes it.

Enumerate  $G_1 = \{h_1, h_2, \dots\}$  and fix a stage  $k \geq 1$ . For each  $i \leq k$ : the finitely many products  $U_{g,n}^{(i)} U_{h,n}^{(i)}$  and  $U_{gh,n}^{(i)}$  with  $g, h \in \{h_1, \dots, h_k\}$  lift the same corona elements, so there is  $n_0$  with all the corresponding defects at most  $1/k$  for  $n \geq n_0$ ; and since  $\limsup_{n \rightarrow \infty} \|U_{g_i,n}^{(i)} - 1\| = 2\varepsilon_i$ , there are

infinitely many  $n \in \mathbb{N}$  with  $\|U_{g_i, n}^{(i)} - 1\| \geq \varepsilon_i$ . Choose  $n_i(k) \in \mathbb{N}$  satisfying both conditions and set

$$\Phi_k(g) = \bigoplus_{i \leq k} U_{g, n_i(k)}^{(i)} \in \mathcal{U}(M_{D_k}(\mathbb{C})), \quad D_k = \sum_{i \leq k} d_{n_i(k)}^{(i)}.$$

Define  $\Theta_\infty(g)$  as the class of  $(\Phi_k(g))_{k \in \mathbb{N}}$  in  $\mathcal{U}_{\text{cor}}((D_k)_{k \in \mathbb{N}})$ . For fixed  $g, h \in G_1$ , both lie in  $\{h_1, \dots, h_k\}$  for all large  $k$ , and the operator norm of a block sum is the maximum of the block norms, so  $\|\Phi_k(g)\Phi_k(h) - \Phi_k(gh)\| \leq 1/k \rightarrow 0$ ; together with  $\Phi_k(1) = 1$  this makes  $\Theta_\infty$  a corona representation of  $G_1$ . Consequently  $\Theta_\infty(R) = 1$ . Conversely, if  $g \notin R$ , pick  $i$  with  $g \in g_i R$ ; then  $\Theta_\infty(g) = \Theta_\infty(g_i)$ , and for every  $k \geq i$ ,  $\|\Phi_k(g_i) - 1\| \geq \|U_{g_i, n_i(k)}^{(i)} - 1\| \geq \varepsilon_i$ , so the representative of  $\Theta_\infty(g_i)$  stays at least  $\varepsilon_i$  from the identity and  $\Theta_\infty(g) \neq 1$ . Hence  $\ker \Theta_\infty = R$ , proving (1), and  $\Theta_\infty$  descends to an injective corona representation of  $G_1/R$ , proving (2). For the final equivalence: if  $G_1$  is MF, an injective corona representation forces  $R = \{1\}$ ; if  $R = \{1\}$ , then (2) says  $G_1 = G_1/R$  is MF.  $\square$

**Corollary 9.4** (exact radical from a candidate quotient). *Let  $N \trianglelefteq G_1$  satisfy  $N \leq \text{Rad}_{\text{MF}}(G_1)$ . Every corona representation of  $G_1$  factors uniquely through  $G_1/N$ . If in addition  $G_1/N$  is MF, then*

$$\text{Rad}_{\text{MF}}(G_1) = N.$$

*Thus proving that a concrete defect subgroup is invisible and that its quotient is MF computes both the entire MF radical and the largest MF quotient.*

*Proof.* The containment  $N \leq \ker \Theta$  gives the unique quotient factorization for every corona representation  $\Theta$ . For the reverse containment in the displayed equality, compose the quotient map  $G_1 \rightarrow G_1/N$  with an injective corona representation of  $G_1/N$ : its kernel is exactly  $N$ , while the MF radical lies in the kernel of every corona representation.  $\square$

## 9.2. Targets beyond coronas.

**Corollary 9.5** (subalgebras of coronas). *Let  $H$ ,  $(\Lambda, \iota, t, c)$  and  $F \subseteq N_{\text{comp}}$  be as in Theorem 7.2. Let  $A$  be a nonzero  $C^*$ -subalgebra of a norm matrix corona  $\mathcal{Q}$  whose unit  $p := 1_A$  is not necessarily the unit of  $\mathcal{Q}$ , and let  $\Theta: H \rightarrow \mathcal{U}(A)$  be a group homomorphism. Then  $\Theta(f) = 1_A$  for every  $f \in F$ . The same conclusion holds when  $A$  is a unital separable MF algebra — for instance a unital separable quasidiagonal algebra (Fact 2.4), in particular an AF algebra, or  $C_{\text{red}}^*(F_k)$  for a free group  $F_k$  [HT, Corollary 8.4].*

*Proof.* Every unitary  $u$  of  $A$  satisfies  $u = pup$  and  $u^*u = uu^* = p$ , so  $\mathcal{U}(A) \subseteq \mathcal{U}(p\mathcal{Q}p)$ , and  $p \neq 0$ . Lift  $p$  to coordinate projections  $p_n$  and discard the zero coordinates. The same zero-extension argument used for bounded products identifies  $p\mathcal{Q}p$  unitally with the norm matrix corona of the nonzero corners  $p_n M_{d_n}(\mathbb{C}) p_n \cong M_{\text{rank } p_n}(\mathbb{C})$ . Composing,  $\Theta$  becomes a corona representation of  $H$ , and Theorem 7.2 gives  $\Theta(f) = 1_A$  for  $f \in F$ . If

$A$  is a unital separable MF algebra, fix a  $*$ -monomorphism  $\psi: A \rightarrow \mathcal{Q}_1$  into a norm matrix corona, unital or not [BK]; then  $\psi(A)$  is a  $C^*$ -subalgebra of  $\mathcal{Q}_1$  with unit  $\psi(1_A)$ , and applying the first part to  $\psi \circ \Theta$  gives  $\psi(\Theta(f)) = \psi(1_A)$ , whence  $\Theta(f) = 1_A$  by injectivity. Unital separable quasidiagonal algebras embed unitaly into coronas by Fact 2.4, AF algebras are quasidiagonal, and  $C_{\text{red}}^*(F_k)$  is MF for every  $k \in \mathbb{N} \cup \{\infty\}$  by [HT, Corollary 8.4].  $\square$

**Corollary 9.6.** *The group  $E$  admits no injective homomorphism into the unitary group of any  $C^*$ -subalgebra of any norm matrix corona — in particular, none into  $\mathcal{U}(A)$  for  $A$  a unital separable MF algebra, a unital separable quasidiagonal algebra, or  $C_{\text{red}}^*(F_k)$ .*

*Proof.* By the proof of Theorem A, the pattern  $(\Gamma, \iota, t, c)$  and  $F = \{1, w\}$  satisfy the hypotheses of Corollary 9.5 in  $H = E$ ; hence any such homomorphism kills  $w$ , while  $w \neq 1$  by Proposition 4.3.  $\square$

### 9.3. Stable finiteness does not imply MF.

**Lemma 9.7.** *Let  $G_1$  be a countable discrete group.*

- (1) *The canonical tracial state  $\tau$  on  $C_{\text{red}}^*(G_1)$  is faithful.*
- (2) *Every unital  $C^*$ -algebra  $A$  with a faithful tracial state is stably finite: for every nonempty finite index set  $I$ , every isometry in  $M_I(A)$  is a unitary. In particular this holds in  $M_k(A)$  for every  $k \geq 1$ .*

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*Proof.* (1) Realize  $C_{\text{red}}^*(G_1) \subseteq B(\ell^2 G_1)$  as the norm closure of the span of the left translation unitaries  $\lambda_g$ , and let  $\rho_g$  be the right translation unitaries,  $(\rho_g \xi)(h) = \xi(hg)$ ; each  $\rho_g$  commutes with each  $\lambda_{g'}$ , hence with every element of  $C_{\text{red}}^*(G_1)$ . The canonical trace is  $\tau(x) = \langle x\delta_e, \delta_e \rangle$ . If  $\tau(x^*x) = 0$  then  $\|x\delta_e\| = 0$ , so  $x\delta_e = 0$ , and then  $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$  for every  $g \in G_1$ . The vectors  $\delta_g$  are total in  $\ell^2 G_1$ , so  $x = 0$ .

(2) On  $M_k(A)$  the functional  $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$  is a state with the trace property  $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$ , and it is faithful: the  $i$ -th diagonal entry of  $x^*x$  is  $\sum_j x_{ji}^*x_{ji}$ , so  $\tau_k(x^*x) = 0$  forces  $\tau(x_{ji}^*x_{ji}) = 0$ , hence  $x_{ji} = 0$ , for all  $i, j$ . If  $v \in M_k(A)$  satisfies  $v^*v = 1$ , then  $y := 1 - vv^*$  is a projection with  $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$ ; since  $y = y^*y$ , faithfulness gives  $y = 0$ , i.e.  $vv^* = 1$ .  $\square$

*Proof of Theorem D.*  $C_{\text{red}}^*(E)$  is unital and separable since  $E$  is countable; by Lemma 9.7 it carries a faithful tracial state and is stably finite. It is not an MF algebra by Theorem A.  $\square$

**Remark 9.8.** Blackadar and Kirchberg proved that MF algebras are stably finite [BK]. The general converse was already refuted as a consequence of the negative solution to the Connes embedding problem [JNVWY], [Go23, Corollary 5.2]; Theorem D instead realizes that failure as the reduced group  $C^*$ -algebra of an explicit finitely presented group, with a faithful trace. The example is not nuclear:  $\Gamma$  embeds into  $E$  (Corollary 4.5); the infinite group  $\Gamma$  has property (T) and is therefore not amenable, since amenable groups with

property (T) are compact, hence finite when discrete [BHV, Theorem 1.1.6]; and the reduced  $C^*$ -algebra of a nonamenable group is not nuclear [BO, Theorem 2.6.8]. The nuclear territory is untouched: whether every stably finite separable *nuclear*  $C^*$ -algebra is quasidiagonal remains open, and by [TWW] every faithful trace on a separable nuclear  $C^*$ -algebra satisfying the UCT is quasidiagonal.

#### 9.4. The ambient HNN extension.

**Lemma 9.9** (residually finite groups are MF). *Every countable residually finite group is MF.*

*Proof.* Let  $G_1$  be countable and residually finite; if  $G_1$  is trivial the statement is trivially true, so enumerate its nontrivial elements  $x_1, x_2, \dots$  (with repetitions if  $G_1$  is finite). For each  $k$  choose a finite-index normal subgroup  $N_k \trianglelefteq G_1$  with  $x_k \notin N_k$ , put  $M_k = N_1 \cap \dots \cap N_k$ , a finite-index normal subgroup, and let  $\varphi_k: G_1 \rightarrow G_1/M_k$  be the quotient map onto the finite group  $G_1/M_k$ . Let  $\pi_k$  be the left regular representation of  $G_1/M_k$  composed with  $\varphi_k$ , a finite-dimensional unitary representation of  $G_1$ . Define  $\Theta(g)$  as the class of  $(\pi_k(g))_k$  in  $\mathcal{U}_{\text{cor}}((|G_1/M_k|)_k)$ ; each  $\pi_k$  is an exact representation, so  $\Theta$  is a corona representation. For  $g = x_i$  and  $k \geq i$  the element  $\varphi_k(x_i)$  is nontrivial, so the permutation matrix  $\pi_k(x_i)$  moves some standard basis vector  $\delta$ , giving  $\|(\pi_k(x_i) - 1)\delta\| = \sqrt{2}$  and  $\|\pi_k(x_i) - 1\| \geq \sqrt{2}$ . Hence the representatives of  $\Theta(x_i)$  stay at distance at least  $\sqrt{2}$  from the identity, and  $\Theta$  is injective.  $\square$

The following  $C^*$ -algebraic order observation is independent of the coordinate proof of Theorem 7.2; it is used only for the maximal group  $C^*$ -algebra below.

**Lemma 9.10** (order identity). *Let  $H_1$  be a countable group, let  $\Lambda_1$  have property (T), let  $\iota_1: \Lambda_1 \rightarrow H_1$  be a homomorphism, and suppose  $t_1 \iota_1(\Lambda_1) t_1^{-1} \subseteq \iota_1(\Lambda_1)$ . If  $\bar{p} = (\iota_1)_*(p_{\Lambda_1}) \in C_{\max}^*(H_1)$ , then*

$$u_{\iota_1(\lambda)} \bar{p} = \bar{p} \quad (\lambda \in \Lambda_1), \quad \bar{p} \leq u_{t_1} \bar{p} u_{t_1}^*.$$

*Proof.* The first identity is the image of  $u_\lambda p_{\Lambda_1} = p_{\Lambda_1}$ . For the second, evaluate in an arbitrary unitary representation  $\pi$  of  $H_1$ . By Fact 2.2,  $\pi(\bar{p})$  projects onto the  $\iota_1(\Lambda_1)$ -fixed vectors. If  $\xi$  is fixed and  $\eta = \pi(t_1)^{-1} \xi$ , then, for every  $\lambda$ , choose  $\lambda'$  with  $t_1 \iota_1(\lambda) t_1^{-1} = \iota_1(\lambda')$ ; one obtains

$$\pi(\iota_1(\lambda)) \eta = \pi(t_1)^{-1} \pi(\iota_1(\lambda')) \xi = \eta.$$

Thus the fixed space is contained in its  $\pi(t_1)$ -translate, which is the range of  $\pi(u_{t_1} \bar{p} u_{t_1}^*)$ . A faithful representation gives the asserted order in  $C_{\max}^*(H_1)$ .  $\square$

**Proposition 9.11** (the ambient group). *Let  $G$  be the ascending HNN extension (7).*



- (1)  $C_{\max}^*(G)$  contains an explicitly described nonunitary isometry. In particular  $C_{\max}^*(G)$  is not finite, a fortiori not stably finite, and not an MF algebra.
- (2)  $G$  is residually finite, hence MF.

*Proof.* (1) Write  $\iota_G: \Gamma \rightarrow G$  for the canonical homomorphism and  $\bar{p} = (\iota_G)_*(p_\Gamma) \in C_{\max}^*(G)$  for the image of the Kazhdan projection. The data  $(G, \Gamma, \iota_G, t)$  satisfy the hypotheses of Lemma 9.10, because  $t \iota_G(\gamma) t^{-1} = \iota_G(\alpha(\gamma))$ ; hence  $\bar{p} \leq Q_1 := u_t \bar{p} u_t^*$ . Put

$$s = \bar{p} u_t^* + (1 - Q_1).$$

From  $\bar{p} \leq Q_1$  we get  $\bar{p} Q_1 = Q_1 \bar{p} = \bar{p}$ , hence  $(1 - Q_1) \bar{p} = 0$  and  $\bar{p} (1 - Q_1) = 0$ ; also  $Q_1 u_t \bar{p} = u_t \bar{p} u_t^* u_t \bar{p} = u_t \bar{p}$ , so  $(1 - Q_1) u_t \bar{p} = 0$  and, taking adjoints,  $\bar{p} u_t^* (1 - Q_1) = 0$ . Therefore

$$\begin{aligned} s^* s &= u_t \bar{p} u_t^* + u_t \bar{p} (1 - Q_1) + (1 - Q_1) \bar{p} u_t^* + (1 - Q_1) \\ &= Q_1 + (1 - Q_1) = 1 \end{aligned}$$

and

$$\begin{aligned} s s^* &= \bar{p} u_t^* u_t \bar{p} + \bar{p} u_t^* (1 - Q_1) + (1 - Q_1) u_t \bar{p} + (1 - Q_1) \\ &= \bar{p} + (1 - Q_1) = 1 - (Q_1 - \bar{p}) : \end{aligned}$$

$s$  is an isometry with defect  $1 - s s^* = Q_1 - \bar{p}$ . Thus  $s^* s = 1$  makes  $s$  left-invertible, with left inverse  $s^*$ , whereas  $s s^* \neq 1$  makes  $s$  a nonunit. Equivalently, with  $a = s^*$  and  $b = s$  one has  $ab = 1$  but  $ba \neq 1$ , so  $C_{\max}^*(G)$  is not directly finite. LEAN 4 exact

It remains to check  $Q_1 \neq \bar{p}$ . Let  $H_0 = t \iota_G(\Gamma) t^{-1} \leq G$  and let  $\pi$  be the quasi-regular representation of  $G$  on  $\ell^2(G/H_0)$ , with basepoint  $b = H_0 \in G/H_0$ . As in the proof of Lemma 9.10,  $\pi(\bar{p})$  is the orthogonal projection onto the  $\iota_G(\Gamma)$ -invariant vectors, and  $\pi(Q_1) = \pi(t) \pi(\bar{p}) \pi(t)^*$ . The vector  $\pi(t)^{-1} \delta_b = \delta_{t^{-1} H_0}$  is  $\iota_G(\Gamma)$ -invariant: for  $\gamma \in \Gamma$ ,

$$\iota_G(\gamma) t^{-1} H_0 = t^{-1} (t \iota_G(\gamma) t^{-1}) H_0 = t^{-1} H_0,$$

since  $t \iota_G(\gamma) t^{-1} \in H_0$ ; hence  $\pi(Q_1) \delta_b = \delta_b$ . On the other hand  $\pi(\iota_G(a)) \delta_b = \delta_{a H_0} \neq \delta_b$ : if  $a H_0 = H_0$  then  $a \in H_0$ , and applying the homomorphism  $G \rightarrow \bar{G}$  of Lemma 4.2, which restricts to the identity on  $\Gamma$  and sends  $t$  to  $D$ , would give  $a \in D \Gamma D^{-1} = \alpha(\Gamma)$  inside  $\text{GL}_4(\mathbb{Z}[1/2])$ , contradicting Lemma 3.1. So  $\delta_b$  is not  $\iota_G(\Gamma)$ -invariant, i.e.  $\pi(\bar{p}) \delta_b \neq \delta_b$  while  $\pi(Q_1) \delta_b = \delta_b$ ; hence  $Q_1 \neq \bar{p}$  and  $s$  is not a unitary.

If  $C_{\max}^*(G)$  embedded unittally into a norm matrix corona, the image of  $s$  would be an isometry there, hence a unitary by Lemma 6.3(1), and injectivity would force  $s s^* = 1$ , a contradiction; so  $C_{\max}^*(G)$  is not an MF algebra, and the isometry  $s$  itself witnesses the failure of finiteness.

(2)  $G$  is the mapping torus of the injective endomorphism  $\alpha$  of the finitely generated linear group  $\Gamma \leq \text{GL}_4(\mathbb{Z})$ , so  $G$  is residually finite by Borisov–Sapir [BoS, Theorem 1.6]; the MF property follows from Lemma 9.9. We note that the proof of [BoS, Theorem 1.6] depends on Hrushovski’s work on the elementary theory of the Frobenius automorphism, which remains



unpublished; this external input affects only the present proposition, not Theorem A.  $\square$

**Remark 9.12.** Proposition 9.11 separates group-level and  $C^*$ -level MF behavior in the direction opposite to Theorem D:  $G$  is an MF group whose maximal group  $C^*$ -algebra is not even finite, while  $E$  is a non-MF group whose reduced group  $C^*$ -algebra is stably finite. The one-way implication — if  $C_{\max}^*$  or  $C_{\text{red}}^*$  of a group is an MF algebra then the group is MF — is recorded in [Sh].

### 9.5. Failure of permanence properties.

**Corollary 9.13** (quotients). *MF groups are not closed under quotients: the free group  $F_8$  is MF, and  $E$  is a quotient of  $F_8$ .*

*Proof.*  $E$  is generated by eight elements (8), hence is a quotient of  $F_8$ . Free groups are residually finite:  $F_8$  embeds into  $F_2$  (a subgroup of index 7 of  $F_2$  is free of rank 8 by the classical Nielsen–Schreier theorem),  $F_2$  embeds into  $\text{SL}_2(\mathbb{Z})$  by Sanov’s classical ping-pong pair  $(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}), (\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix})$ , residual finiteness passes to subgroups, and  $\text{GL}_2(\mathbb{Z})$  is residually finite by reduction modulo primes (a nontrivial integer matrix stays nontrivial modulo any prime exceeding the absolute values of the entries of its difference from the identity). So  $F_8$  is MF by Lemma 9.9, while  $E$  is not MF by Theorem A.  $\square$

**Proposition 9.14** (the witness group is non-MF). *Let  $W = \text{CLamp}(\bar{X}) \rtimes \bar{G}$  be the witness group of Proposition 4.3. Then  $W$  is finitely generated, the homomorphism  $E \rightarrow W$  of that proof is surjective, and the central sign  $z = -1 \in \text{CLamp}(\bar{X})$  satisfies  $z \in \text{Rad}_{\text{MF}}(W) \setminus \{1\}$ . In particular  $W$  is not MF.*

*Proof.* The image of the homomorphism  $E \rightarrow W$  contains the six base generators of  $\Gamma$ , hence  $\Gamma$ ; it contains  $D$ ; so it contains  $\bar{G} = \langle \Gamma, D \rangle$ . It contains  $c_o$ , hence  $c_{g\Gamma} = g c_o g^{-1}$  for every  $g \in \bar{G}$ , which is all of  $\{c_x : x \in \bar{X}\}$  since  $\bar{G}$  acts transitively on  $\bar{X} = \bar{G}/\Gamma$ . Finally  $(c_{D\Gamma} c_{aD\Gamma})^2 = -c_{D\Gamma}^2 c_{aD\Gamma}^2 = -1 = z$  because the two sites are distinct (Lemma 4.2), so the image contains  $z$  and therefore all of  $\text{CLamp}(\bar{X}) = \langle \pm 1, c_x : x \in \bar{X} \rangle$ : the map is surjective, and  $W$  is generated by the images of the eight generators of  $E$ . The proof of Proposition 4.3 sends  $w$  to  $z$ , so Lemma 9.2 gives  $z \in \text{Rad}_{\text{MF}}(W)$ , and  $z \neq 1$  in  $W$  by Construction 4.1. A group whose MF radical is nontrivial admits no injective corona representation.  $\square$

**Proposition 9.15** (the marked word is tracially visible). *There are a countable sofic group  $S$ , a Connes-embeddable finite tracial von Neumann algebra  $\mathcal{M}$ , and homomorphisms*

$$\rho: E \longrightarrow S, \quad \pi: E \longrightarrow \mathcal{U}(\mathcal{M})$$

*such that*

$$\rho(w) \neq 1, \quad \pi(w) = -1.$$

Consequently, if  $\text{Rad}_{\text{hyp}}(E)$  and  $\text{Rad}_{\text{sof}}(E)$  denote the corresponding intersections of kernels over hyperlinear and sofic targets, then

$$w \in \text{Rad}_{\text{MF}}(E) \setminus (\text{Rad}_{\text{hyp}}(E) \cup \text{Rad}_{\text{sof}}(E)).$$

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*Proof.* For  $n \geq 1$  put

$$X_n = \Gamma / \alpha^n(\Gamma), \quad X = \bigsqcup_{n \geq 1} X_n.$$

Every  $X_n$  is finite. The formulas

$$F(g\alpha^n(\Gamma)) = \alpha(g)\alpha^{n+1}(\Gamma)$$

define an injective map  $F: X \rightarrow X$  satisfying  $F(\gamma x) = \alpha(\gamma)F(x)$ . Let  $A_{\text{Cl}}$  be the tracial complex CAR algebra generated by self-adjoint Majoranas  $C_x$ ,  $x \in X$ , with  $C_x C_y = -C_y C_x$  for  $x \neq y$ . The permutation action of  $\Gamma$  on  $X$  is trace preserving, and

$$C_x \mapsto C_{F(x)}, \quad u_\gamma \mapsto u_{\alpha(\gamma)}$$

extends to a trace-preserving injective endomorphism  $\eta$  of

$$(*) \quad N = A_{\text{Cl}} \rtimes \Gamma.$$

We first check Connes embeddability rather than assume it. Let  $A_m$  be the finite-dimensional Clifford algebra on  $X_1 \sqcup \cdots \sqcup X_m$  and put  $N_m = A_m \rtimes \Gamma$ . The action on these finitely many coordinates has finite image; its kernel  $\Lambda_m$  has finite index in the residually finite linear group  $\Gamma$  and acts trivially on  $A_m$ . Thus  $N_m$  is a finite twisted crossed product of  $A_m \bar{\otimes} L(\Lambda_m)$  and is Connes embeddable. The  $N_m$  increase trace preservingly to  $N$ , so  $N$  is Connes embeddable.

Let  $(\tilde{N}, \tilde{\eta})$  be the minimal automorphic dilation of  $(N, \eta)$ . It is an increasing tracial union of copies of  $N$ , hence is Connes embeddable. The crossed product

$$\mathcal{M} = \tilde{N} \rtimes_{\tilde{\eta}} \mathbb{Z}$$

is Connes embeddable as well: this is the standard amenable-crossed-product permanence, obtained here directly by Følner compression for  $\mathbb{Z}$  followed by matrix microstates for  $\tilde{N}$ . Let  $U$  be the implementing unitary, with  $UxU^* = \tilde{\eta}(x)$ .

Take  $x_0 = \alpha(\Gamma) \in X_1$ , put  $a_0 = C_{x_0}$ , and define

$$\pi(\gamma) = u_\gamma \quad (\gamma \in \Gamma), \quad \pi(t) = U, \quad \pi(c) = U^* a_0 U.$$

The HNN relations follow from  $(*)$ . Moreover

$$U[U^* a_0 U, u_\gamma]U^* = [a_0, u_{\alpha(\gamma)}] = 0,$$

because  $\alpha(\Gamma)$  fixes  $x_0$ . Thus the image of  $c$  is an involution centralizing  $\Gamma$ . Finally,

$$\pi(tct^{-1}) = C_{x_0}, \quad \pi(a(tct^{-1})a^{-1}) = C_{ax_0}.$$

Since  $a \notin \alpha(\Gamma)$ , the two sites are distinct. Their Majoranas anticommute, so their group commutator is the scalar  $-1$ . This proves  $\pi(w) = -1$ ; scalarity also verifies all defining centrality and order-two relations for  $w$ .

There is also a group-valued version of the same tower. With  $Y_m = X_1 \sqcup \cdots \sqcup X_m$ , put

$$K_m = \text{ClLamp}(Y_m) \rtimes \Gamma, \quad K = \bigcup_m K_m = \text{ClLamp}(X) \rtimes \Gamma.$$

The finite Clifford lamp group on  $Y_m$  makes  $\Gamma$  a finite-index subgroup of  $K_m$ . Hence every  $K_m$  is sofic, and so is the directed union  $K$ . The maps  $F$  and  $\alpha$  define an injective endomorphism of  $K$ . Its automorphic direct limit  $\widetilde{K}$  is again sofic. Finally,

$$S = \widetilde{K} \rtimes \langle T \rangle,$$

where  $T$  implements that automorphism, is sofic because its normal subgroup  $\widetilde{K}$  is sofic and its quotient  $\mathbb{Z}$  is amenable. We use here the standard permanence of soficity under subgroups, direct limits, finite extensions, and extensions by amenable quotients [ES].

Let  $z$  be the central sign of the Clifford lamp group. The assignments

$$\rho(\gamma) = \gamma \quad (\gamma \in \Gamma), \quad \rho(t) = T, \quad \rho(c) = T^{-1}c_{x_0}T$$

satisfy the literal presentation by the same covariance calculation as above, and the two Majoranas in the marked commutator are distinct. Thus  $\rho(w) = z \neq 1$ . Notice that this argument uses the valid permanence of soficity under directed unions and extensions by an amenable *quotient*; it does not assert closure for arbitrary amenable-by-sofic extensions.

The countable image  $\pi(E)$  is hyperlinear because  $\mathcal{M}$  is Connes embeddable, and the quotient map  $E \rightarrow \pi(E)$  detects  $w$ . Hence  $w$  is not in the hyperlinear radical. The map  $\rho$  shows that it is not in the sofic radical. Theorem A puts it in the MF radical.  $\square$

**Corollary 9.16** (sofic does not imply MF). *There is an explicit finitely generated sofic group which is not MF. Namely, the image  $Q = \rho(E) \leq S$  in Proposition 9.15 is sofic, while  $1 \neq \rho(w) \in \text{Rad}_{\text{MF}}(Q)$ .*

*Proof.* Subgroups of sofic groups are sofic, and  $Q$  is generated by the images of the eight displayed generators of  $E$ . Portability (Lemma 9.2) sends  $w \in \text{Rad}_{\text{MF}}(E)$  to  $\rho(w) \in \text{Rad}_{\text{MF}}(Q)$ . This element is nontrivial by Proposition 9.15, so  $Q$  is not MF.  $\square$

**Corollary 9.17** (a simple sofic group with full MF radical). *There is a countable simple sofic group  $S_{\text{sof}}$  with*

$$\text{Rad}_{\text{MF}}(S_{\text{sof}}) = S_{\text{sof}}.$$

*Thus  $S_{\text{sof}}$  is hyperlinear, but every homomorphism from it into an MF group is trivial.*

*Proof.* Let  $Q$  be the finitely generated sofic non-MF group from Corollary 9.16. Elek–Szabó prove that every countable sofic group embeds in a countable simple sofic group [ES, Theorem 1]; choose an embedding  $j: Q \hookrightarrow S_{\text{sof}}$ . Portability sends the nontrivial element  $\rho(w) \in \text{Rad}_{\text{MF}}(Q)$  to a nontrivial element of  $\text{Rad}_{\text{MF}}(S_{\text{sof}})$ . The latter is normal, so simplicity makes it the whole group. Sofic groups are hyperlinear. Finally, if  $f: S_{\text{sof}} \rightarrow M$  and  $M$  is MF, then an injective corona representation of  $M$  composed with  $f$  is trivial; hence  $f$  is trivial.  $\square$

**Corollary 9.18** (extensions). *MF groups are not closed under extensions, even split ones: in the split exact sequence*

$$1 \longrightarrow \text{CLamp}(\bar{X}) \longrightarrow W \longrightarrow \bar{G} \longrightarrow 1,$$

*the kernel is locally finite, hence amenable and MF; the quotient  $\bar{G} \leq \text{GL}_4(\mathbb{Z}[1/2])$  is a finitely generated linear group, hence residually finite by Mal'cev's theorem [Ma] and MF by Lemma 9.9; and  $W$  is not MF (Proposition 9.14). In particular there is a finitely generated non-MF group that is (locally finite)-by-(residually finite). This contrasts with the positive permanence theory for MF groups in [Sh].*

*Proof.* Exactness and splitting hold by construction of the semidirect product. Every finite subset of  $\text{CLamp}(\bar{X})$  lies in the subgroup generated by  $\pm 1$  and finitely many  $c_x$ , a finite 2-group, as in the proof of Theorem C; so  $\text{CLamp}(\bar{X})$  is locally finite, hence amenable, hence MF by Facts 2.3 and 2.4. The remaining assertions were proved above.  $\square$

**Corollary 9.19** (a simple envelope). *There exists a 2-generator simple group  $S$  with  $\text{Rad}_{\text{MF}}(S) = S$ : every corona representation of  $S$ , and every homomorphism from  $S$  into any MF group, is trivial.*

*Proof.* By the embedding theorem of Goryushkin [Go] and Schupp [Sch], proved independently, the countable group  $E$  embeds into a 2-generator simple group  $S$ . By Lemma 9.2 applied to the embedding, the image of  $w$  is a nontrivial element of  $\text{Rad}_{\text{MF}}(S)$ . The radical is a normal subgroup of the simple group  $S$ , so  $\text{Rad}_{\text{MF}}(S) = S$ , i.e. every corona representation of  $S$  is trivial. If  $\varphi: S \rightarrow M$  is a homomorphism to an MF group and  $j$  an injective corona representation of  $M$ , then  $j \circ \varphi$  is a corona representation, hence trivial, so  $\varphi$  is trivial. The simple-envelope theorem belongs to Goryushkin and Schupp; what is added here is the nontrivial image of the radical element.  $\square$

**9.6. Finitely presented Kazhdan envelopes.** We use the following recent external embedding theorem. Chatterji and Kassabov prove that, given a finitely generated recursively presented group  $H$  and  $1 \neq h \in H$ , there is an embedding  $j: H \hookrightarrow \Lambda$  into a finitely presented property (T) group  $\Lambda$  which is normally generated by  $j(h)$  [CK, Theorem 3]. The arXiv record [arXiv:2601.22907v1](https://arxiv.org/abs/2601.22907v1) was submitted on January 30, 2026; the manuscript itself is dated February 2, 2026. We use only that theorem and make no claim that its construction is part of the present work.

**Theorem 9.20** (a finitely presented Kazhdan group with full MF radical). *There exist a finitely presented property (T) group  $\Lambda$  and an embedding  $j: E \hookrightarrow \Lambda$  such that*

$$\text{Rad}_{\text{MF}}(\Lambda) = \Lambda.$$

*Every corona representation of  $\Lambda$ , and every homomorphism from  $\Lambda$  to an MF group, is trivial. Moreover every finite-dimensional linear representation of  $\Lambda$ , over every field, is trivial, and  $\Lambda$  is perfect.*

*Proof.* Apply [CK, Theorem 3] to  $(E, w)$ . Portability gives  $j(w) \in \text{Rad}_{\text{MF}}(\Lambda)$ . The radical is normal, whereas  $j(w)$  normally generates  $\Lambda$ , so the radical is all of  $\Lambda$ . The statements about corona representations and maps to MF groups follow from Proposition 9.3.

Let  $\pi: \Lambda \rightarrow \text{GL}(V)$  with  $V$  finite-dimensional over an arbitrary field. Theorem B, applied to  $\pi \circ j$ , gives  $\pi(j(w)) = 1$ ; normal generation gives  $\pi = 1$ . Finally  $w$  is a commutator, hence  $j(w)$  lies in the commutator subgroup of  $\Lambda$ . That normal subgroup contains a normal generator, so it is all of  $\Lambda$  and  $\Lambda$  is perfect.  $\square$

The four supporting formal declarations isolate the normal-closure and full-radical deductions used above. They do not formalize the external Chatterji–Kassabov existence theorem or package its output into an unconditional formal witness.

In particular  $\Lambda$  has no nontrivial finite or residually finite quotient. It also has no nontrivial amenable quotient and no nontrivial finitely generated linear quotient: quotients of the finitely generated group  $\Lambda$  are countable, amenable groups are MF by Facts 2.3 and 2.4, and finitely generated linear groups are residually finite by Mal’cev’s theorem [Ma]. Thus  $\Lambda$  is minimally almost periodic. By the classical Peter–Weyl theorem [PW], its Bohr compactification is trivial; this last Peter–Weyl implication is not part of the formal companion.

**Theorem 9.21** (a universal Kazhdan group with full MF radical). *There exists a finitely presented property (T) group  $\Lambda_{\text{univ}}$  which contains every finitely presented group as a subgroup and satisfies*

$$\text{Rad}_{\text{MF}}(\Lambda_{\text{univ}}) = \Lambda_{\text{univ}}.$$

*Every homomorphism from  $\Lambda_{\text{univ}}$  to an MF group is trivial.*

*Proof.* Enumerate all finite group presentations and form the free product of the groups they present. The Higman–Neumann–Neumann and Higman embedding theorems [HNN, Hig] embed this recursively presented countable group into a finitely presented group  $U$ . Thus  $U$  contains every finitely presented group. Choose  $i: E \hookrightarrow U$ . Portability and injectivity give  $1 \neq i(w) \in \text{Rad}_{\text{MF}}(U)$ . Apply [CK, Theorem 3] to  $(U, i(w))$ . Its image in the resulting finitely presented property (T) envelope normally generates, so portability forces the final MF radical to be the entire group.

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Universality survives the embedding of  $U$ , and the last assertion follows from Proposition 9.3.  $\square$

Likewise, the supporting declarations verify only the full-radical and homomorphism deductions once a normally generated envelope is supplied. Neither the effective HNN–Higman construction nor the Chatterji–Kassabov existence theorem is asserted as a Lean axiom.

The directionality is worth emphasizing:  $\Lambda_{\text{univ}}$  contains every finite group and every finitely presented residually finite, amenable, or linear group, while admitting no nontrivial quotient in any of those MF classes.

**Corollary 9.22** (opposite full and reduced behavior in one group). *For the same universal Kazhdan group,*

$$C_{\text{red}}^*(\Lambda_{\text{univ}}) \text{ is stably finite but non-MF,}$$

*whereas  $C_{\text{max}}^*(\Lambda_{\text{univ}})$  contains a proper isometry and is therefore neither finite nor stably finite.*

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*Proof.* The canonical trace on  $C_{\text{red}}^*(\Lambda_{\text{univ}})$  is faithful, so the proof of Lemma 9.7, applied to every finite matrix amplification, gives stable finiteness. If the reduced algebra embedded faithfully in a norm matrix corona, restriction to its left-regular unitaries would give an injective corona representation of  $\Lambda_{\text{univ}}$ , contradicting its full MF radical.

The finitely presented ambient HNN group  $G$  of Proposition 9.11 embeds in  $\Lambda_{\text{univ}}$ . For  $H \leq K$ , the canonical map  $C_{\text{max}}^*(H) \rightarrow C_{\text{max}}^*(K)$  is injective: induce any unitary representation of  $H$  to  $K$  and recover the original representation on the identity-coset summand, so the maximal norm is preserved. Thus the explicit proper isometry in  $C_{\text{max}}^*(G)$  remains proper in  $C_{\text{max}}^*(\Lambda_{\text{univ}})$ . A proper isometry witnesses failure of finiteness and stable finiteness.  $\square$

**9.7. The scaling family.** For an integer  $m \geq 2$  let  $E_m$  be the group with the presentation (8), except that the three relations  $tv_it^{-1} = v_i^2$  are replaced by

$$tv_it^{-1} = v_i^m \quad (1 \leq i \leq 3);$$

thus  $E = E_2$ , while the marked word  $w_m = [tct^{-1}, v_1(tct^{-1})v_1^{-1}]$  is unchanged.

**Theorem 9.23** (scaling family). *For every  $m \geq 2$ , the group  $E_m$  is finitely presented,  $w_m$  is a nontrivial central involution in  $E_m$ , and every corona representation of  $E_m$  kills  $w_m$ . Consequently  $E_m$  is not MF, neither  $C_{\text{max}}^*(E_m)$  nor  $C_{\text{red}}^*(E_m)$  is an MF algebra, and  $C_{\text{red}}^*(E_m)$  is stably finite but not MF.*

*Proof.* Finite presentation is clear from the displayed presentation.

*The pattern.* Define  $\alpha_m: \Gamma \rightarrow \Gamma$  by  $\alpha_m(v, A) = (mv, A)$ . The computation of Lemma 3.1 applies with 2 replaced by  $m$ :  $\alpha_m$  is an injective endomorphism with image  $m\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$ , and  $a = (e_1, 1) \notin \alpha_m(\Gamma)$  because  $e_1 \notin m\mathbb{Z}^3$ . On the six base generators  $\alpha_m$  fixes  $x, y, z$  and sends  $v_i$  to  $v_i^m$ , so the relations of  $E_m$  assert  $t\gamma t^{-1} = \alpha_m(\gamma)$  on generators; hence, with  $\iota_m: \Gamma \rightarrow E_m$  the

canonical homomorphism,  $t\iota_m(\gamma)t^{-1} = \iota_m(\alpha_m(\gamma))$  for all  $\gamma \in \Gamma$ , and  $c$  commutes with  $\iota_m(\Gamma)$ : the tuple  $(\Gamma, \iota_m, t, c)$  is a marked Kazhdan pattern in  $E_m$  (Fact 2.1). As in the proof of Theorem A,  $d^2 = tc^2t^{-1} = 1$  in  $E_m$ , and Lemma 8.1 gives  $w_m = [d, \iota_m(a)]^2 \in N_{\text{comp}}$ ; the subgroup  $\{1, w_m\}$  is finite and normal, so Theorem 7.2 kills  $w_m$  in every corona representation.

*Nontriviality.* Let  $D_m = \text{diag}(m, m, m, 1) \in \text{GL}_4(\mathbb{Z}[1/m])$ . As in Lemma 4.2,  $D_m \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} D_m^{-1} = \begin{pmatrix} A & mv \\ 0 & 1 \end{pmatrix}$  is the matrix of  $\alpha_m(v, A)$ , and  $D_m^{-1}aD_m = \begin{pmatrix} 1 & \frac{1}{m}e_1 \\ 0 & 1 \end{pmatrix} \notin \Gamma$ , so in  $\bar{G}_m := \langle \Gamma, D_m \rangle \leq \text{GL}_4(\mathbb{Z}[1/m])$  the cosets  $D_m\Gamma$  and  $aD_m\Gamma$  of  $\bar{X}_m := \bar{G}_m/\Gamma$  are distinct. The proof of Proposition 4.3 applies with  $(\bar{G}, \bar{X}, D)$  replaced by  $(\bar{G}_m, \bar{X}_m, D_m)$  — its only inputs were the conjugation identity, the fixing of the base coset by  $\Gamma$ , and the coset separation — and sends  $w_m$  to  $[c_{D_m\Gamma}, c_{aD_m\Gamma}] = -1 \neq 1$  in  $\text{Cilamp}(\bar{X}_m) \rtimes \bar{G}_m$ .

The final assertions follow exactly as in the proofs of Theorems A and D, with  $E_m$  in place of  $E$ .  $\square$

**Remark 9.24.** We do not claim the groups  $E_m$  are pairwise non-isomorphic; multiplicity is addressed in Corollary 9.26. For every  $d \geq 3$  the same construction runs with  $\mathbb{Z}^d \rtimes \text{SL}_d(\mathbb{Z})$  in place of  $\Gamma$  — property (T) holds by the same references [BHV, Sections 1.4–1.5 and 4.2] — once a finite presentation of the base is fixed; we confine the formal statement to  $d = 3$ , where the presentation (6) is completely explicit.

### 9.8. Multiplicity.

**Lemma 9.25.** *Every subgroup of an MF group is MF; equivalently, a group containing a non-MF subgroup is not MF.*

*Proof.* The restriction of an injective corona representation to a subgroup is an injective corona representation.  $\square$

**Corollary 9.26** (multiplicity).

- (1) *The groups  $E \times \mathbb{Z}^k$ ,  $k \in \mathbb{N}$ , are pairwise non-isomorphic, finitely presented, and non-MF.*
- (2) *There exist  $2^{\aleph_0}$  pairwise non-isomorphic finitely generated non-MF groups.*

*Proof.* Every group in sight contains an isomorphic copy of  $E$ , hence is non-MF by Theorem A and Lemma 9.25.

(1)  $E \times \mathbb{Z}^k$  is finitely presented because  $E$  is. Its abelianization is  $E^{\text{ab}} \times \mathbb{Z}^k$ ; since  $E$  is finitely generated,  $E^{\text{ab}}$  is a finitely generated abelian group, of some torsion-free rank  $r_0 < \infty$ , so the abelianization of  $E \times \mathbb{Z}^k$  has torsion-free rank  $r_0 + k$ , an isomorphism invariant that separates distinct  $k$ .

(2) By a classical construction of B. H. Neumann [Neu] there is a family  $(N_i)_{i \in I}$  with  $|I| = 2^{\aleph_0}$  of pairwise non-isomorphic 2-generator groups. Put  $G_i = E \times N_i$ , a finitely generated non-MF group. For a fixed countable group  $G_1$ , the set  $\{i \in I : G_i \cong G_1\}$  is countable: a countable group has only countably many finitely generated subgroups (each is determined by a



finite generating subset), hence countably many isomorphism classes of such; if  $G_i \cong G_1$  then  $N_i$  embeds into  $G_1$  as  $\{1\} \times N_i$ ; and the  $N_i$  are pairwise non-isomorphic, so  $i \mapsto [N_i]$  is injective on the set in question. A map from a set of cardinality  $2^{\aleph_0}$  with countable fibres has image of cardinality  $2^{\aleph_0}$ ; apply this to  $i \mapsto [G_i]$ .  $\square$

**9.9. Algorithmic unrecognizability.** We use the Adian–Rabin theorem in its standard Markov-property form [Ad58, Rab]; an English translation of Adian’s original articles, with a modern statement of the theorem, is [NB]. The form used here: if an isomorphism-invariant property has a finitely presented positive example and a finitely presented forbidden group which cannot embed in any finitely presented group having the property, then the property is undecidable from finite presentations. All effectivity assertions below are relative to a fixed standard Gödel coding of finite presentations; the accompanying formal statements are reductions parameterized by such a coding and by the Adian–Rabin test-group transformation, which is not itself formalized.

**Theorem 9.27** (MF recognition is undecidable). *There is no algorithm which, given a finite group presentation, decides whether the presented group is MF.*

*Proof.* Being MF is inherited by subgroups (Lemma 9.25). The trivial group is a finitely presented positive example, while the finitely presented non-MF group  $E$  cannot embed in any MF group. Thus MF is a Markov property, and Adian–Rabin applies.  $\square$

**Theorem 9.28** (negative instances are not recursively enumerable). *For any standard effective coding of finite group presentations, the set of codes presenting non-MF groups is not recursively enumerable. Consequently there is no sound and complete recursively verifiable system of finite certificates for failure of the MF property in finitely presented groups.*

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The two Lean declarations are reduction theorems parameterized by a concrete primitive-recursive presentation coding and a proof-carrying ‘AdianRabinReduction’. They do not construct that transformation. The classical proof below supplies the paper-level external input; the badges are therefore conditional rather than exact formal certifications of the effective statements.

*Proof.* Fix a finitely presented group  $N$  with undecidable word problem. The effective test-group construction in the proof of Adian–Rabin associates to every word  $u$  a finite presentation  $P_u$  such that  $u = 1$  in  $N$  implies  $G(P_u) = 1$ , whereas  $u \neq 1$  implies that  $E$  embeds in  $G(P_u)$ . Hence  $G(P_u)$  is MF exactly when  $u = 1$ . Trivial words are recursively enumerable. If non-MF finite presentations were also recursively enumerable, pulling that enumeration back along  $u \mapsto P_u$  would enumerate the nontrivial words of  $N$ ; running both enumerations in parallel would decide its word problem, a contradiction.



The certificate statement follows because the objects accepted by a recursive finite verifier form a recursively enumerable set.  $\square$

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### 9.10. Uniform invisibility.

**Theorem 9.29** (uniform invisibility). *Let  $H, (\Lambda, \iota, t, c)$  and  $F \subseteq N_{\text{comp}}$  be as in Theorem 7.2, and fix  $f_0 \in F$  — for instance  $H = E$  and  $f_0 = w$ . For every  $\varepsilon > 0$  there exist  $\delta > 0$  and a finite subset  $F_0 \subseteq H$  such that for every  $r \geq 1$  and every map  $\varphi: H \rightarrow \mathcal{U}(M_r(\mathbb{C}))$  with*

$$\|\varphi(g)\varphi(h) - \varphi(gh)\| \leq \delta \quad \text{for all } g, h \in F_0,$$

*one has  $\|\varphi(f_0) - 1\| < \varepsilon$ .*

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*Proof.* Suppose the statement fails for some  $\varepsilon_0 > 0$ . Fix an enumeration  $H = \{h_1, h_2, \dots\}$  (with repetitions if  $H$  is finite) and put  $F_k = \{h_1, \dots, h_k\}$ . For each  $k \geq 1$ , the failure at  $(\delta, F_0) = (1/k, F_k)$  produces  $r_k \geq 1$  and  $\varphi_k: H \rightarrow \mathcal{U}(M_{r_k}(\mathbb{C}))$  with  $\|\varphi_k(g)\varphi_k(h) - \varphi_k(gh)\| \leq 1/k$  for all  $g, h \in F_k$ , but  $\|\varphi_k(f_0) - 1\| \geq \varepsilon_0$ . Define  $\Theta(g)$  as the class of the sequence  $(\varphi_k(g))_{k \in \mathbb{N}}$  in  $\mathcal{Q}((r_k)_{k \in \mathbb{N}})$ , a unitary element. For fixed  $g, h \in H$  we have  $g, h \in F_k$  for all large  $k$ , so the displayed defect tends to 0 and  $\Theta(g)\Theta(h) = \Theta(gh)$ . Taking  $g = h = 1$  gives  $\|\varphi_k(1)^2 - \varphi_k(1)\| \leq 1/k$  for large  $k$ , hence  $\|\varphi_k(1) - 1\| \leq 1/k$  after multiplying by the unitary  $\varphi_k(1)^*$ ; so  $\Theta(1) = 1$ , and then  $\Theta(g)\Theta(g^{-1}) = \Theta(1) = 1$  for every  $g$ . Thus  $\Theta$  is a corona representation of  $H$ , and Theorem 7.2 gives  $\Theta(f_0) = 1$ , i.e.  $\|\varphi_k(f_0) - 1\| \rightarrow 0$  — contradicting  $\|\varphi_k(f_0) - 1\| \geq \varepsilon_0$ .  $\square$

**Remark 9.30.** Conversely, the uniform statement applied to unitary lifts recovers the kill part of Theorem 7.2, so the two are equivalent. The compactness proof yields no explicit  $(\delta, F_0)$ , and none is claimed; an effective version would require tracking the Kazhdan constant of  $\Lambda$  through Sections 7–8, which we do not pursue.

The closedness assertion below is a standard consequence of Dadarlat’s finite-local characterization of MF groups and his observation that groups locally embeddable in MF groups are MF [Dad, Proposition 3.2 and Remark 3.3(i)], together with the standard fact that a marked limit locally embeds in its approximating groups; for a modern formulation see [FFGS, Theorem 2.16]. We claim no novelty for that implication. We record it because the word-ball certificate is useful here.

**Theorem 9.31** (marked-space closedness and the literal cylinder). *For every  $k$ , the MF locus in the space  $\mathcal{G}_k$  of  $k$ -marked groups is closed; equivalently, the non-MF locus is open. More precisely, every non-MF marked group has a finite reduced-word ball whose relation pattern defines a neighborhood consisting entirely of non-MF groups.*

*For the canonical eight-generator marking of  $E$ , there is the explicit clopen neighborhood*

$$\mathcal{O}_E = \{(H; h_1, \dots, h_8) : r(h_1, \dots, h_8) = 1 \ (r \in R), \quad w(h_1, \dots, h_8) \neq 1\}.$$

Every group in  $\mathcal{O}_E$  is non-MF. With the literal relator set of Definition 3.2, membership is determined by the relation pattern on reduced words of length at most 34.

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*Proof.* The tensor-power separation amplifier gives the following normalized local criterion: a countable group is MF exactly when, for every finite subset and every positive error, it has an operator-norm unitary model whose distinct elements are separated by at least 1.

Suppose  $(G_n, S_n) \rightarrow (G, S)$  and every  $G_n$  is MF. Given a finite  $F \subseteq G$ , choose words in the free marked generators representing every element of  $F \cup F^2$ . On a sufficiently small cylinder, all equalities and inequalities among these finitely many words agree with those in  $G$ . A normalized local model in one such  $G_n$  therefore pulls back, word by word, to a local model of  $G$  with the same multiplicative error and separation. The criterion makes  $G$  MF. This proves closedness; recording the finitely many control words inside one reduced word ball gives the stronger finite-radius statement.

For the literal neighborhood, the relators give a surjection  $E \twoheadrightarrow H$ . If the marked word survives in  $H$ , portability of  $w \in \text{Rad}_{\text{MF}}(E)$  puts its nontrivial image in  $\text{Rad}_{\text{MF}}(H)$ , so  $H$  is non-MF. Equality or inequality of each fixed word is clopen, and there are finitely many relators. Direct reduction of the displayed words gives length at most 16 for  $w$  and at most 34 for every relator (the longest are the centrality relators  $[w, g]$ ), proving the asserted radius.  $\square$

**9.11. A torsion-free full-radical group.** We now combine Theorem 7.3 with the torsion-free small-cancellation architecture of Fournier-Facio [FF]. This proves Theorem E and removes the former torsion-free existence question.

Use the input groups from [FF, Section 2]: a universal finitely presented torsion-free group  $U$ , a finitely presented infinite simple torsion-free group  $S$ , and a torsion-free hyperbolic property-(T) group  $H$ . The group  $U$  embeds in a finitely presented torsion-free property-(T) group  $P$ . Universality gives an embedding

$$(15) \quad P_1 \times P_2 \times S \leq P, \quad P_1 \cong P_2 \cong P.$$

For chosen isomorphisms  $\phi_i: P \rightarrow P_i$ , form the double ascending HNN extension

$$(16) \quad \mathcal{E} = \langle P, u_1, u_2 \mid u_i p u_i^{-1} = \phi_i(p) \quad (p \in P, i = 1, 2) \rangle.$$

Fournier-Facio proves that  $\mathcal{E}$  is finitely presented, torsion-free, and acylindrically hyperbolic [FF, Section 2].

Fix  $1 \neq s \in S$  and put

$$\Lambda = P, \quad \iota: P \hookrightarrow \mathcal{E}, \quad t = u_1, \quad c = u_1^{-1} s u_1.$$

**Lemma 9.32** (defect saturation). *These data form a marked Kazhdan pattern in  $\mathcal{E}$ . Its compression defect is*

$$(17) \quad D = \langle\langle [s, p] : p \in P \rangle\rangle_{\mathcal{E}},$$

and  $D$  contains  $S$  and a suitable subgroup of  $\mathcal{E}$ .

*Proof.* The relation  $u_1 P u_1^{-1} = P_1 \leq P$  is the one-sided compression condition. For  $p \in P$ , the element  $u_1 p u_1^{-1}$  lies in  $P_1$ , which commutes with  $S$  by (15); hence

$$[u_1^{-1} s u_1, p] = u_1^{-1} [s, u_1 p u_1^{-1}] u_1 = 1.$$

Thus  $c$  centralizes  $P$ , while  $t c t^{-1} = s$ , proving (17).

The infinite simple group  $S$  is nonabelian and has trivial center. Choose  $r \in S$  with  $[s, r] \neq 1$ . Then  $1 \neq [s, r] \in D \cap S$ , and  $D \cap S \trianglelefteq S$ ; simplicity gives  $S \leq D$ .

Let  $N = \langle\langle S \rangle\rangle_{\mathcal{E}} \leq D$ . This is an infinite normal subgroup of the acylindrically hyperbolic group  $\mathcal{E}$ , so it acts non-elementarily by Osin's normal-subgroup lemma [Osi, Lemma 7.1]. Torsion-freeness excludes every nontrivial finite subgroup. Hull's suitable-subgroup criterion therefore supplies independent loxodromic elements in  $N$  generating a suitable subgroup [Hul, Corollary 5.7 and Lemma 5.8].  $\square$

The following relative form of Hull's common-quotient argument is the routing step needed below.

**Lemma 9.33** (defect-routing common quotient). *Let  $A$  and  $B$  be finitely presented torsion-free acylindrically hyperbolic groups, with  $B$  having property (T). Suppose  $D \trianglelefteq A$  contains a suitable subgroup. For every finite  $F_A \subset A$ , there are epimorphisms*

$$A \xrightarrow{\alpha} Q \xleftarrow{\beta} B$$

such that:

- (1)  $Q$  is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic;
- (2)  $Q$  has property (T);
- (3)  $\alpha(D) = Q$ ;
- (4)  $\alpha$  is injective on  $F_A$ .

*Proof.* Set  $F_0 = A * B$ . Inside the prescribed suitable subgroup of  $D$ , choose a two-generated suitable subgroup  $L = \langle a_1, a_2 \rangle$ . Choose independent loxodromic elements  $b_1, b_2 \in B$ . The free factors are hyperbolically embedded in  $F_0$ , and transitivity of hyperbolic embedding gives

$$\{\langle a_1 \rangle, \langle a_2 \rangle, \langle b_1 \rangle, \langle b_2 \rangle\} \hookrightarrow_h F_0.$$

Apply Hull's Theorem 7.1 with suitable subgroup  $L$ , with a finite generating set of  $B$  as the target elements, and with a protected ball containing  $F_A$  and

the four selected loxodromic elements. It gives an epimorphism  $\gamma: F_0 \twoheadrightarrow F_1$  such that

$$(18) \quad \gamma(B) \leq \gamma(L) \leq \gamma(D) \leq \gamma(A),$$

$F_1$  is acylindrically hyperbolic, and  $\gamma$  is injective on  $F_A$ . The strengthened choice used in the proof of Hull's Corollary 7.4 preserves the two cyclic subgroups generated by the  $b_i$  as an independent hyperbolically embedded pair. Thus  $\gamma(B)$  remains suitable in  $F_1$  [Hul, proof of Corollary 7.4]. From (18),

$$F_1 = \langle \gamma(A), \gamma(B) \rangle = \gamma(A).$$

Choose a two-generated suitable subgroup  $T = \langle x, y \rangle$  inside  $\gamma(B)$ . Apply Hull's Theorem 7.1 a second time, routing a finite generating set of  $F_1$  into  $T$  and protecting  $\gamma(F_A)$ . The resulting epimorphism  $\eta: F_1 \twoheadrightarrow Q$  satisfies  $Q = \eta(T)$ , remains injective on  $\gamma(F_A)$ , and has acylindrically hyperbolic target. Hence  $Q$  is generated by two elements. Moreover,

$$T \leq \gamma(B) \leq \gamma(D) \implies \eta\gamma(D) = Q.$$

Therefore  $\alpha = \eta\gamma|_A$  and  $\beta = \eta\gamma|_B$  are both epimorphisms,  $\alpha(D) = Q$ , and  $\alpha$  is injective on  $F_A$ .

Property (T) passes through the epimorphism  $\beta$ . At each stage, Hull's torsion clause says that an element of finite order in the quotient is the image of an element of the same order in the preceding group; since  $A * B$  is torsion-free, so is  $Q$ . Both applications use finitely many target elements and add finitely many relators, so  $Q$  is finitely presented. This is the finite-presentation observation used in [FF, Section 2].  $\square$  LEAN 4 SUPPORT

*Proof of Theorem E.* Apply Lemma 9.33 with

$$A = \mathcal{E}, \quad B = H, \quad D \text{ as in (17)}, \quad F_A = \{1, s\}.$$

Lemma 9.32 supplies the suitable subgroup. We obtain epimorphisms

$$\pi: \mathcal{E} \twoheadrightarrow Q, \quad \rho: H \twoheadrightarrow Q,$$

where  $Q$  has every asserted geometric and finiteness property,  $\pi(s) \neq 1$ , and  $\pi(D) = Q$ .

The marked Kazhdan pattern descends to  $(P, \pi\iota, \pi(u_1), \pi(c))$ . Since  $\pi$  is onto, its compression defect is the image of the original defect, hence is all of  $Q$ . The group  $Q$  itself is a normal property-(T) subgroup of  $Q$ , so Theorem 7.3, applied with  $K = Q$ , gives

$$Q \leq \text{Rad}_{\text{MF}}(Q) \leq Q.$$

Thus  $\text{Rad}_{\text{MF}}(Q) = Q$ , and  $Q$  is nontrivial because  $\pi(s) \neq 1$ .

It remains to verify that the quotient retains Fournier-Facio's nonsocicity configuration. The restriction  $\pi|_S$  is injective: its kernel is normal in the simple group  $S$ , and it is not all of  $S$  because  $\pi(s) \neq 1$ . Put

$$\Gamma' = \pi(P), \quad t_i = \pi(u_i), \quad J = t_1^{-1}\pi(S)t_1.$$

Then  $Q = \langle \Gamma', t_1, t_2 \rangle$  and  $t_i \Gamma' t_i^{-1} \leq \Gamma'$ . The group  $\Gamma'$  is infinite and has property (T),  $[\Gamma', J] = 1$ , and  $t_1 J t_1^{-1} = \pi(S) \leq \Gamma'$ .

We claim  $\Gamma' \cap J = 1$ . After conjugating by  $t_1$ , this is equivalent to

$$(19) \quad \pi(P_1) \cap \pi(S) = 1.$$

Let  $K = (P_1 \times S) \cap \ker \pi$ . If  $(p, z) \in K$  and  $x \in S$ , normality gives

$$[(p, z), (1, x)] = (1, [z, x]) \in K \cap S.$$

Injectivity of  $\pi|_S$  implies  $[z, x] = 1$  for every  $x \in S$ , and hence  $z = 1$  because  $Z(S) = 1$ . If  $\pi(p) = \pi(z')$  with  $p \in P_1$  and  $z' \in S$ , then  $(p, (z')^{-1}) \in K$ , so the preceding argument gives  $z' = 1$  and then  $\pi(p) = 1$ . This proves (19).

Fournier-Facio's criterion [FF, Proposition 1.2] now applies to  $\Gamma' \leq Q$  and  $J$ . If  $Q$  were sofic, then  $J \cong S$  would be LEF. But a finitely presented LEF group is residually finite, whereas an infinite simple group is not residually finite. Therefore  $Q$  is nonsofic.  $\square$

**Corollary 9.34** (quotient-hereditary failure of MF). *For every quotient map  $q: Q \twoheadrightarrow R$ ,*

$$\text{Rad}_{\text{MF}}(R) = R.$$

*Consequently every nontrivial quotient of  $Q$  is non-MF. In particular, the MF group  $F_2$  has a finitely presented torsion-free non-MF quotient, so MF groups are not closed under quotients even within finitely presented torsion-free groups.*

*Proof.* Functoriality gives

$$R = q(Q) = q(\text{Rad}_{\text{MF}}(Q)) \leq \text{Rad}_{\text{MF}}(R) \leq R.$$

The final assertion follows because  $Q$  is two-generated and hence is a quotient of  $F_2$ , while free groups are residually finite and therefore MF.  $\square$

## 10. THE CYCLIC-BASE COMPARISON

We prove Theorem C. The broader phenomenon that a marked central involution can be invisible to every exact finite-dimensional representation but visible to approximate representations is central to Slofstra's work [Slo17]; the point here is its occurrence in the same operator-norm marked-compression pattern as Theorem A. Recall the setting:  $G_0 = \text{BS}(1, 2)$  with base  $\Gamma_0 = \langle \gamma_0 \rangle \cong \mathbb{Z}$  and stable letter  $t$  satisfying  $t\gamma_0 t^{-1} = \gamma_0^2$ ;  $X_0 = G_0/\Gamma_0$ ;  $E_0 = \text{Cilamp}(X_0) \rtimes G_0$  with central sign  $z_0 = -1 \in \text{Cilamp}(X_0)$  as in Construction 4.1. Figure 6 summarizes the comparison.

*Proof of Theorem C.*  $z_0 \neq 1$ : immediate from Construction 4.1, since  $\text{Cilamp}(X_0) \leq \text{Cl}(X_0)^\times$  and  $\text{Cl}(X_0) \neq 0$ .

*Amenability:* every finitely generated subgroup of  $\text{Cilamp}(X_0)$  is contained in the subgroup generated by  $\pm 1$  and finitely many  $c_x$ , which is a finite 2-group (it has order at most  $2^{k+1}$  for  $k$  sites, since every element is  $\pm a$  a reduced product of distinct generators). Hence  $\text{Cilamp}(X_0)$  is locally finite, so amenable. The group  $G_0 = \text{BS}(1, 2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$  is metabelian, so

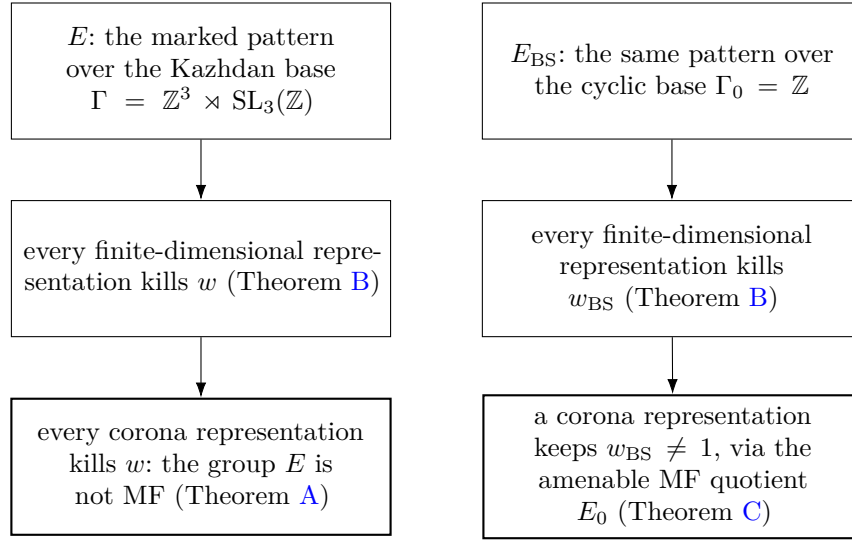


FIGURE 6. The cyclic-base comparison. The finite-dimensional obstruction does not see the base, while the corona conclusion holds exactly on the side where property (T) supplies the uniform fixed-space projection.

amenable. As an extension of the amenable  $G_0$  by the amenable  $\text{Cilamp}(X_0)$ , the group  $E_0$  is amenable.

*MF:* by Facts 2.3 and 2.4, the amenable group  $E_0$  is MF, and an MF embedding  $E_0 \hookrightarrow \text{U}_{\text{cor}}((d_n))$  is injective, so the image of  $z_0$  is nontrivial; under  $\kappa_{(d_n)}$  it is a nontrivial self-adjoint unitary in  $\mathcal{Q}$ : the central sign survives.

*The quotient from the literal presentation:* assign  $\gamma_0 \mapsto \gamma_0$ ,  $t \mapsto t$ , and  $c \mapsto c_o$ , where  $o = \Gamma_0 \in X_0$ . The point  $o$  is fixed by  $\Gamma_0$ , so  $[c_o, \gamma_0] = 1$ . Also  $t\Gamma_0 t^{-1} = \langle \gamma_0^2 \rangle \subseteq \Gamma_0$ . Moreover  $\gamma_0 t \Gamma_0 \neq t \Gamma_0$  in  $G_0$ : otherwise  $t^{-1} \gamma_0 t \in \Gamma_0$ ; in the faithful affine model  $\text{BS}(1, 2) = \mathbb{Z}[1/2] \rtimes \mathbb{Z}$  with  $\gamma_0 = (1, 0)$  and  $t = (0, 1)$  acting by multiplication by 2, we compute  $t^{-1} \gamma_0 t = (\frac{1}{2}, 0) \notin \Gamma_0 = \mathbb{Z} \times \{0\}$ . Consequently

$$w_{\text{BS}} \mapsto [c_{t\Gamma_0}, c_{\gamma_0 t \Gamma_0}] = z_0.$$

The image is central of order two, so every relator of  $E_{\text{BS}}$  is respected. The resulting homomorphism is surjective:  $\gamma_0, t$  generate  $G_0$ , the conjugates of  $c_o$  give all  $c_x$  because  $G_0$  acts transitively on  $X_0$ , and their commutators contain  $z_0$ . Composing this quotient with the MF embedding of  $E_0$  gives the asserted corona representation of  $E_{\text{BS}}$ .

*The finite-dimensional obstruction persists:*  $E_0$  contains the pattern of Theorem B with  $H = E_0$ ,  $\Gamma = \Gamma_0$ ,  $t = t$ ,  $c = c_o$  ( $o = \Gamma_0 \in X_0$ , fixed by  $\Gamma_0$ ), and  $a = \gamma_0 \in \Gamma_0$ . We have  $t\Gamma_0 t^{-1} = \langle \gamma_0^2 \rangle \subseteq \Gamma_0$  and  $[c_o, \Gamma_0] = 1$ . Thus Theorem B says every finite-dimensional representation of  $E_0$ , over every field, kills  $z_0$ . Applying the same theorem directly to the displayed

presentation of  $E_{\text{BS}}$  shows that every finite-dimensional representation kills  $w_{\text{BS}}$ .  $\square$

**Remark 10.1.** Theorem C shows that the argument cannot be generalized merely by deleting its fixed-space rigidity input. The elementary Theorem B is genuinely weaker than the corona conclusion:  $E_0$  has MF approximations even though its finite-dimensional representations all kill the sign.

## 11. LIMITATIONS: HILBERT–SCHMIDT METRICS AND HYPERLINEARITY

The proof of Theorem A does not begin from a merely tracial approximation, for a structural reason worth recording.

Suppose one replaces the norm corona  $\mathcal{Q}$  by a tracial ultraproduct  $\mathcal{M} = \prod^\omega (M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$ , with defects measured in  $\|\cdot\|_2$  rather than operator norm, and asks whether a hyperlinear-style embedding of  $E$  must kill  $w$ . The first coordinate step already breaks: Hilbert–Schmidt multiplicative defects do not control the operator norms of the off-diagonal blocks cut out by the Reynolds spectral projections. Consequently principal compression need not remain multiplicative in operator norm, its Gram defects need not vanish in operator norm, and the dimension-free polar-repair estimate used in Lemma 7.9 is unavailable. Likewise, the finite-stage Kazhdan argument needs operator-norm spectral leakage estimates before it passes to normalized Hilbert–Schmidt norm. A tracial ultraproduct retains only the latter information, so the equal-rank coordinate flip cannot be invoked. Proposition 9.15 shows that the failure is realized sharply for the marked conclusion: there is an exact Connes-embeddable tracial representation with  $w = -1$ . Thus no universal tracial analogue of the MF kill theorem can hold for this word. The proposition does not make the representation faithful, so it decides neither hyperlinearity of  $E$  nor the sofic-versus-hyperlinear problem.

The dividing line in the present argument is thus: *operator-norm matrix models form a stably finite world, in which property (T) compression is rigid; for tracial matrix models the available ambient algebra of the conjugation action can be properly infinite, so the same projection comparison need not collapse to equality.*

This failure is closely related to a current structural question rather than to a known impossibility theorem. Alekseev–Thom ask whether, for a Kazhdan representation into a tracial ultraproduct of matrices, its commutant can be recovered as an ultraproduct of finite-dimensional subalgebras, possibly as coordinate centralizers of suitable lifts [AT, Open Problem 6.2]. Such a coordinate model would provide finite-level structure absent from the argument above, but their question is open. Moreover, any strengthened coordinate-nesting statement which, combined only with the fixed Clifford block in the  $w = -1$  sector, forced  $w = 1$  would contradict Proposition 9.15; some additional hypothesis would be essential.



## TOOL AND COMPUTATIONAL RESOURCE DISCLOSURE

The mathematics in this paper and the accompanying machine-checked development were produced with extensive use of large language models, including Anthropic Claude and OpenAI Codex models in multiple concurrent interactive sessions. Literature search was assisted by web-connected model sessions.

## ACKNOWLEDGEMENTS

[Removed for anonymous review.]

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